

Original inverse-sector quotients: source shifts, heat and covariance angles

Split-Zero research programme

21 September 2026
SZ-20260921-013

The original inverse-power space can be removed through explicit maps of the observation complex, with its complete minimum metric and source relations retained. The kernel determinant changes by $o(kq)$ on the stated original degree regime; multiplication by y^r identifies the result with a full polynomial source at the displayed shifted cutoff.

The coefficient of the retained observed inverse-power determinant is exactly the older C_∂ . The equality below preserves all elliptic constants. A new bound controls the quotient angle under the full source covariance by its actual eigenvalues:

$$|\delta(t) - \delta(0)| \leq \sum_{j=1}^{\min(m,r,\text{rank } \Omega)} \log(1 + t\lambda_j(G(0)^{1/2}\Omega G(0)^{1/2})).$$

For the actual scalar adjacent source, the logarithmic factor is evaluated as $\log(\nu_{N+1-q}/\omega_{N+1}) = O(q)$, giving $o(kq)$. An already activated two-column path retains its own initial angle; the finite change estimate does not replace that baseline.

The residual observed heat outside the removed sector has the squared small-angle bound displayed in IQR8-10. The proper-source and complex current maps retain their full minima and phase data. Their unevaluated scalar values and the full kernel allocation remain part of the active mathematical calculation.

The reader contains every new proof, the complete received continuation, the independent finite quotient derivation and the independent covariance angle derivation. Complete earlier proof sources and the original human source TeX accompany it.

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Part I

Original-metric analytic receivers and new angle estimate

The received continuation supplies an explicit quotient of the original observation complex by its controlled inverse-power sector. This note proves the analytic receiving estimates in its original metrics and adds a rank-sensitive bound for the angle change under the complete source covariance. It also identifies exactly the coefficient already called C_∂ in the older endpoint work. The full received argument and its independent finite verification accompany this note.

1 Objects, guards and the original coefficient

Keep $0 < \delta < 1/2$, $\gamma > 2$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, the original full polynomial $Q_k(y)$, $E = \mathbb{C}[y]/(Q_k)$, $M[p] = [yp]$, and the four cutoffs $N = q-1, q, 2q-1, 2q$. The physical coordinate is $S = k/2 + iy$. The source is $\mu_k = w_h^{*k}$ and G_N is its attained metric. Keep the original onto observation $\Lambda : E \rightarrow B$, its full kernel K , and its fixed coefficient frame I_K . On the proved five-orbit period domain $m = \dim K = 8k - 16$; no larger period domain is asserted.

For $1 \leq r < q$ put $U_r d = \sum_{j=1}^r d_j [y^{-j}]$, $V_r = \text{im } U_r$. Use every finite guard of IVO1-26 and IFH1-21 [1, 2]. In particular, on the eventual regime $r \log(q+2) = o(q)$,

$$H_{r,N}^B \succeq e^{-\varepsilon_{k,r}} H_{r,N}^E, \quad \varepsilon_{k,r} = O_{h,A}((k+Jr) \log(q+Jr+2)) = o(q). \quad (\text{IQR1})$$

Here $H^E = U_r^* G_N U_r$, $H^B = U_r^* \Lambda^* Q_{B,N} \Lambda U_r$ and $Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}$. The exact positive finite IVO factor may replace the exponential bound.

The coefficient in IFH10 is exactly the earlier outer coefficient. Here is the complete dictionary, retaining its scale. At the EIQ potential $\pi\sqrt{x} - 2\log x$, let u, v be the actual positive square-root endpoints and write

$$uK(\kappa) = 2, \quad vE(\kappa) = 4, \quad a = u/v, \quad \kappa^2 = 1 - a^2, \quad w = (u+v)/2, \quad z = uv, \quad c = (v^2 - u^2)/4.$$

ECL1-17 [3] proves $L = \int \log x d\rho_1 = 6\log w - 2\log z - 2$ and $C_\partial = 4\log(4/\pi) + 8\log w - 4\log z - 2\log c$. The retained profile has $\psi(0) = \log(4/\pi)$, and S2 gives

$$\psi(1) = \log \frac{\kappa(1+a)}{E(\kappa)^2} = \log \frac{w\sqrt{c}}{4}, \quad J(1) = 2(\log 2 - 1) - L/2.$$

The middle equality follows from $v = 4/E$, $w = v(1+a)/2$ and $\sqrt{c} = v\kappa/2$. Substitution, with every constant retained, proves

$$\begin{aligned} \mathfrak{C} &= 4[\psi(0) - \psi(1) - J(1) - 1] \\ &= 4\log(4/\pi) - 4\log w - 2\log c + 4 + 2L \\ &= 4\log(4/\pi) + 8\log w - 4\log z - 2\log c = C_\partial. \end{aligned} \quad (\text{IQR2})$$

Thus the already reproduced interval $1.353893 < C_\partial < 1.354751$ applies to this same older coefficient. No numerical approximation is used to prove its identity.

2 Full shifted polynomial sources and their finite errors

The original quotient coordinates and attained metric are

$$\pi_r[p] = \mathcal{D}^r \text{rem}_{Q_k}(y^r p), \quad \bar{G}_{r,N} = (\pi_r G_N^{-1} \pi_r^*)^{-1}, \quad \hat{H}_{K,r,N} = (\pi_r I_K)^* \bar{G}_{r,N} (\pi_r I_K).$$

The finite Schur identity in the supplied equations (3)-(9) yields

$$0 \leq \delta_{r,N} := \log \det H_{K,N} - \log \det \hat{H}_{K,r,N} \leq \min(m, r) \varepsilon_{k,r} =: \mathcal{L}_{k,r}. \quad (\text{IQR3})$$

All ranks can be replaced by the actual cross-Gram rank.

Let $\mathcal{P}_L$ be the original physical- y polynomial space with Gram H_L^{src} for μ_k . In $\mathcal{P}_{N+r}$ keep the full column matrix

$$R_{r,N} = [1, y, \dots, y^{r-1}, Q_k, yQ_k, \dots, y^{N+r-q}Q_k], \quad E_{r,N} = y^r I_K.$$

The last equality uses degree-below- q representatives of I_K . The columns of $R_{r,N}$ are independent: a relation between them would make a degree-below- $r < q$ polynomial divisible by Q_k . Their Gram is therefore positive. Completing the full quadratic minimum gives

$$H_{r,N}^{\text{sh}} = E^* H_{N+r}^{\text{src}} E - E^* H_{N+r}^{\text{src}} R (R^* H_{N+r}^{\text{src}} R)^{-1} R^* H_{N+r}^{\text{src}} E. \quad (\text{IQR4})$$

This is the metric of $M^r I_K$ in $E/\mathcal{P}_{<r}$ attained from the cutoff $N+r$. Multiplication by M^r identifies E/V_r with $E/\mathcal{P}_{<r}$, since $M^r V_r = \mathcal{P}_{<r}$. It sends $\pi_r I_K$ to exactly the classes displayed in IQR4.

Put $n_L = \lceil (L+2)/2 \rceil$, $a_L^\Gamma = \sqrt{n_L(n_L+1)}(1 + \sqrt{L+1})$ and

$$\mathfrak{B}_{r,L} = \sqrt{u_k/\ell_{k,L}} (a_L^\Gamma)^r, \quad \mathfrak{A}_{r,N} = \prod_{j=0}^{r-1} C_{\text{mult}}(h)[2(N+j) + \lfloor k/3 \rfloor].$$

S12-16 [6] proves $\|\mathcal{D}^r p\|_\sigma \leq a_L^\Gamma \|p\|_\sigma$ for every $\deg p \leq L$. Applying this r times in the same Gamma space and using NG2 only before and after gives $\|\mathcal{D}^r p\|_{\mu_k} \leq \mathfrak{B}_{r,L} \|p\|_{\mu_k}$. The intermediate degree decreases; a_L^Γ bounds every step. NG3 [5] gives the displayed $\mathfrak{A}$ for multiplication by y^r , including the entire outgoing degree at every step.

For the forward quotient bound multiply the actual minimum source representative by y^r . For the inverse choose the complete minimum representative in the target and apply $\mathcal{D}^r$; the discarded Taylor polynomial belongs to the exact target relation $\mathcal{P}_{<r}$. These are admissible lifts of the inverse isomorphisms, so

$$\mathfrak{B}_{r,N+r}^{-2} \hat{H}_{K,r,N} \preceq H_{r,N}^{\text{sh}} \preceq \mathfrak{A}_{r,N}^2 \hat{H}_{K,r,N}. \quad (\text{IQR5})$$

In particular, with $\log^+ x = \max(0, \log x)$, an explicit error is

$$\left| \mathcal{R} \log \det H_{r,N}^{\text{sh}} - \mathcal{R} \log \det H_{K,N} \right| \leq 2\mathcal{L}_{k,r} + 2m \sum_N \max\{\log^+ \mathfrak{A}_{r,N}, \log^+ \mathfrak{B}_{r,N+r}\} = o(kq). \quad (\text{IQR6})$$

Indeed each logarithmic comparison in IQR5 lies between $-2m \log \mathfrak{B}$ and $2m \log \mathfrak{A}$; IQR3 contributes at most $2\mathcal{L}$ under two positive and two negative signs. NG2-3 gives $\log^+ \mathfrak{A} + \log^+ \mathfrak{B} = O_h(k+r \log q)$ for these full degrees $N+r = O(q)$. This is $o(q)$. No shifted cutoff has been relabelled as an unshifted one.

For $1 \leq s \leq r$, let $Z = (I - P_{V_r,N})I_K$. Its original Gram is $\hat{H}_{K,r,N}$. The exact Taylor-division inverse identity kills its rank- s remainder after projection perpendicular to

V_r , proving $\|M^s x\| \geq \mathfrak{B}_{s,N}^{-1} \|x\|$ for $x \perp V_r$. The original decomposition $M = C_N + R_N$ has rank $R_N = 1$, $\|C_N\| \leq B_h q$, and $\|R_N\| = \epsilon_N$ [7]. The min-max principle on $\ker R_N$ gives $\sigma_j(M) \leq B_h q$ for $j \geq 2$, and the triangle inequality bounds $\sigma_1(M) \leq \epsilon_N + B_h q$. Applying the m th exterior power to M^s therefore proves

$$\begin{aligned} -2m \log \mathfrak{B}_{s,N} &\leq \log \frac{\det((M^s Z)^* G_N M^s Z)}{\det \hat{H}_{K,r,N}} \\ &\leq 2s \log(\epsilon_N + B_h q) + 2s(m-1) \log(B_h q). \end{aligned} \quad (\text{IQR7})$$

Here the product of the largest m singular values is the norm of the m th exterior map; this follows by exteriorizing a singular-value decomposition. Its norm is sub-multiplicative under composition. Since $\log^+ \epsilon_N = O_h(q)$, IQR3 and IQR7 preserve the original kq coefficient for $s = o(k)$. The original source order, not a truncated multiplication, supplies this bound.

3 Quadratic residual heat and the actual eigenclass stack

All adjoints and projections in this section use G_N . For $r \geq 1$, let $W^E = P_{B,N} V_r$ and $P_{\text{rem}} = P_{B,N} - P_{W^E}$. The inclusion $W^E \subset K^\perp$ makes this an orthogonal projection. Its image is $(K + V_r)^\perp$: an element of $K^\perp$ pairs with $P_B V_r$ exactly as it pairs with V_r . The fixed class $u = -Q_k(0)[y^{-1}]$ lies in V_r . Thus $P_{\text{rem}} u = 0$.

Let a_N be a unit smallest right singular vector, and retain the actual finite LS guard $E_N > 2D_N$ and $\eta_N = D_N/(E_N - D_N)$. LS4 gives $\|(I - P_u)a_N\| \leq \eta_N$, and hence $\|P_{\text{rem}} a_N\|^2 \leq \eta_N^2$. In the full orthonormal singular basis the other eigenvalues of $M^\dagger M$ are at least D_N^{-2} . Every corresponding projection weight lies in $[0, 1]$. Therefore

$$0 \leq \text{Tr}(P_{\text{rem}} e^{-\tau M^\dagger M}) \leq \eta_N^2 e^{-\tau s_N^2} + (q-1) e^{-\tau/D_N^2}. \quad (\text{IQR8})$$

This proves the quadratic angle error without discarding a heat direction.

Write $\theta_N = \|P_B u\|^2 / \|u\|^2$ and use the exact IFH11 relative error $r_N(\tau)$. When $r_N(\tau) < 1$, its positive lower bound for the original measured heat gives the finite fraction estimate

$$0 \leq \frac{\text{Tr}(P_{\text{rem}} e^{-\tau M^\dagger M})}{\text{Tr}(P_B e^{-\tau M^\dagger M})} \leq \frac{\eta_N^2 + (q-1) e^{-\tau(D_N^{-2} - s_N^2)}}{\theta_N(1 - r_N(\tau))}. \quad (\text{IQR9})$$

The right side tends to zero at the unchanged common time $\tau_k^* = \exp\{2q[\log(q/k) - 1 - I(\delta, \gamma)/2]\}$, including the two high cutoffs whose surviving mass is very small. In fact $\theta_N \geq e^{-O(k \log q)}$, $\eta_N = e^{-q \log(q/k) + O_h(q)}$ and $\tau_k^*(D_N^{-2} - s_N^2)$ grows faster than every power of q . Summing the four nonnegative upper bounds in IQR8 proves

$$\limsup \frac{\log |\mathcal{R} \text{Tr}(P_{\text{rem}} e^{-\tau_k^* M^\dagger M})|}{q \log(q/k)} \leq -2, \quad \log 0 = -\infty. \quad (\text{IQR10})$$

IQR9 also proves that the removed observed sector carries a fraction tending to one of the full observed heat at every cutoff.

Let $S_r : E/V_r \rightarrow E$ be the original minimum section, an isometry onto $V_r^\perp$. LS2 reads $M^{-1} = B_N - u \varphi_N / Q_k(0)$ with $\|B_N\| \leq D_N$. For $x \perp V_r$, $x = (I - P_{V_r}) B_N M x$, so

$$S_r^\dagger M^\dagger M S_r \succeq D_N^{-2} I. \quad (\text{IQR11})$$

This is the full-output square of the rectangular map MS_r . The block resolvent and memory formula IFH16–17 gives its exact relation to the independently compressed observed action.

For the older NP18 stack, preserve multiplication by the physical coordinate S : $A_S = (k/2)I + iM$. If e_λ is the cardinal y -eigenclass, its physical eigenvalue is $\mu = k/2 + i\lambda$. For every integer depth $d \geq 0$ used at all four cutoffs,

$$\left\| (\Lambda, \Lambda A_S, \dots, \Lambda A_S^d) e_\lambda \right\|^2 = \left(\sum_{j=0}^d |\mu|^{2j} \right) \|\Lambda e_\lambda\|_{Q_{B,N}}^2. \quad (\text{IQR12})$$

Each term follows from $A_S^j e_\lambda = \mu^j e_\lambda$; the output stack has the direct-sum metric with the full $Q_{B,N}$. The geometric factor cancels exactly under $\mathcal{R}$, for growing as well as fixed d . IFH20 bounds the observed fraction $\beta_N \in [e^{-Ck \log(q+2)}, 1]$ on the detected-line domain. Consequently $|\mathcal{R} \log \beta_N| \leq 2Ck \log(q+2) = o(q)$. Together with IQR2, NP19 now has coefficient $C_\partial = \mathfrak{C}$ on that domain. This is an update to that specific canonical angle term.

4 Complete source covariance and a new angle bound

Let the complete actual normal source have Gram H_Z and value map F_Z , with $\ker H_Z \subseteq \ker F_Z$. Retain its full cross terms in $\Omega = F_Z H_Z^+ F_Z^* \succeq 0$. For $t \geq 0$ the specified activation gives $G(t) = (G^{-1} + t\Omega)^{-1}$. At a fixed period $\pi = \pi_r$ is fixed, so its full quotient is exactly

$$\overline{G}(t) = (\pi G^{-1} \pi^* + t\pi \Omega \pi^*)^{-1}. \quad (\text{IQR13})$$

No orthogonality between unprocessed source groups is inserted.

Put $C = G^{1/2} \Omega G^{1/2}$ and $S = \overline{G}(0)^{1/2} \pi G^{-1/2}$. Then $SS^* = I$, and the quotient covariance is $\overline{C} = SC S^*$. The min-max characterization gives $\lambda_j(C) \geq \lambda_j(\overline{C}) \geq \lambda_{j+r}(C)$ in decreasing order. Apply the increasing functions $f_t(x) = tx/(1+tx)$ and $g_t(x) = \log(1+tx)$ and sum. The two sides telescope to at most the top r terms, proving

$$\begin{aligned} 0 &\leq \text{Tr } f_t(C) - \text{Tr } f_t(\overline{C}) \leq r, \\ 0 &\leq \text{Tr } g_t(C) - \text{Tr } g_t(\overline{C}) \leq r \log(1 + t\|C\|). \end{aligned} \quad (\text{IQR14})$$

The complete C7 bound [8] therefore acquires just r/q after division by q , for $b \geq 0$ and $t = e^{-2bq \log(q/k)}$. For the H23 scale $t = e^{-2\theta q}$, $\theta \geq 0$, $\log(1 + \|C\|) = O_h(q \log(q/k))$ gives an additional $o(q^2)$ when $r \log q = o(q)$. The ensuing original observation removes another m dimensions and costs at most $m \log(1 + t\|C\|) = o(q^2)$. This proves the received (48) on its original simple-quartet and period domain, with the same C_B . It does not infer a kq estimate from that q^2 result.

There is, however, a sharper angle control for small covariance rank. Keep the fixed disjoint subspaces $K, V = V_r$ and $S_0 = K + V$. For a subspace X with a fixed full frame I_X , write $P_X(t) = I_X(I_X^* G(t) I_X)^{-1} I_X^* G(t)$. Use the concatenated frame $[I_K, U_r]$ on S_0 , and set

$$\delta(t) = \log \det H_K(t) + \log \det H_V(t) - \log \det H_{S_0}(t) \geq 0.$$

Its nonnegativity and its equality to the source/observed angle loss follow from the full Schur determinant identity. Since $G'(t) = -G(t)\Omega G(t)$, cyclicity of trace proves, for $t > 0$,

$$t\delta'(t) = -\text{Tr}(D_t A_t), \quad D_t = tG(t)^{1/2} \Omega G(t)^{1/2}, \quad A_t = \tilde{P}_K(t) + \tilde{P}_V(t) - \tilde{P}_{S_0}(t), \quad (\text{IQR15})$$

where $\tilde{P}_X = G(t)^{1/2} P_X G(t)^{-1/2}$ is the exact Euclidean orthogonal projection corresponding to the original metric. Thus IQR15 is a proved coordinate identity, with both subspaces moving under this isometry and the physical subspaces fixed.

Here is the spectrum of A_t , including all nontrivial directions. Take orthonormal frames of the two transformed subspaces and a singular value decomposition of their cross Gram. Its singular values are $c_j \in [0, 1)$ because the two original subspaces have zero intersection. For each $c_j > 0$ choose paired unit vectors $e_j \in G(t)^{1/2} K$, $f_j \in G(t)^{1/2} V$ with $\langle e_j, f_j \rangle = c_j$. In the orthonormal pair $e_j, (f_j - c_j e_j)/\sqrt{1 - c_j^2}$, the restriction of A_t is

$$\begin{pmatrix} c_j^2 & c_j \sqrt{1 - c_j^2} \\ c_j \sqrt{1 - c_j^2} & -c_j^2 \end{pmatrix}. \quad (\text{IQR16})$$

Its characteristic polynomial is $\lambda^2 - c_j^2$. All remaining directions give zero, including directions exterior to S_0 . Thus A_t has paired eigenvalues $\pm c_j$, with $\nu(t) = \text{rank}(I_K^* G(t) U_r) \leq \min(m, r)$ pairs.

Set $d_\Omega = \text{rank } \Omega$ and $c = \|C\|$. The positive operator D_t has rank d_Ω and norm $tc/(1+tc)$. To see this exactly, $G(t)\Omega$ is similar to $(I+tC)^{-1}C$, and D_t is similar to $tG(t)\Omega$. If $A_t = A_+ - A_-$ is its positive/negative spectral decomposition, then $0 \preceq A_\pm \preceq I$ and each has rank $\nu(t)$. For any positive D , $\text{Tr}(DA_\pm) \leq \min\{\text{rank } D, \nu(t)\} \|D\|$: use both the bound by $\text{Tr } D$ and the bound by $\nu(t)\|D\|$. It follows that

$$|t\delta'(t)| \leq \min(m, r, d_\Omega) \frac{tc}{1+tc}, \quad (\text{IQR17})$$

$$|\delta(t) - \delta(0)| \leq \sum_{j=1}^{\ell} \log(1 + t\lambda_j(C)) \leq \ell \log(1 + tc), \quad \ell = \min(m, r, d_\Omega).$$

For the stronger middle bound, diagonalize D_t . The diagonal weights of A_+ or A_- in that eigenbasis lie in $[0, 1]$ and their sum is at most $\nu(t)$. Their weighted eigenvalue sum is therefore at most the sum of the largest ℓ eigenvalues of D_t ; moving weight from a smaller eigenvalue to a larger one proves this directly. These eigenvalues are $t\lambda_j(C)/(1 + t\lambda_j(C))$. Integrating proves the displayed spectral sum with its original covariance eigenvalues retained. The integral from zero is finite; smoothness of the positive finite Grams extends the identity to $t = 0$. If $c = 0$, the path is constant.

This supplies an explicit control of the previously retained correction in the received equation (49). At each endpoint,

$$0 \leq \delta_{r,N}(t_N) \leq \min(m, r) \varepsilon_{k,r} + \min(m, r, d_{\Omega,N}) \log(1 + t_N \|C_N\|) =: L_N. \quad (\text{IQR18})$$

Hence its four-sign absolute value is at most $\max\{L_{q-1} + L_q, L_{2q-1} + L_{2q}\}$. If the actual source covariance has rank bounded by d_k , $d_k \log(q+2) = o(k)$, and its retained logarithmic size is $O_h(q \log(q+2))$, this proves that the source and observation quotient still preserves the original activated kernel coefficient through $o(kq)$ when its starting metric is the canonical G_N to which IQR1 applies. The rank-sensitive finite change IQR17 also holds from an already activated positive metric. In that case its initial angle is retained; rank alone does not transfer IQR1 to it. The full joint covariance can have rank q ; IQR18 retains that rank and does not assert a kq conclusion for it.

For the actual adjacent scalar source this size is evaluated, not assumed. Let $\phi_{N+1} = p_{N+1}/\sqrt{\omega_{N+1}}$ be the added orthonormal source column, and $f_N = [\phi_{N+1}]$ its original value. Put $\alpha_N = f_N^* G_N f_N$. The full monic relation norm calculation H3 [7], with $s = N + 1 - q$, gives

$$1 + \alpha_N = \frac{\nu_s}{\omega_{N+1}}, \quad \log(1 + \alpha_N) = 2q\psi(s/q) + O_h(k \log^2(q+2)) = O_h(q). \quad (\text{IQR22})$$

Indeed the original minimum representative of $[p_{N+1}]$ is the difference between its full minimum monic relation and p_{N+1} . The two orthogonal parts have squared norms $\nu_s - \omega_{N+1}$ and ω_{N+1} , respectively. Dividing by the unchanged ω_{N+1} proves IQR22. All previous relations remain in that minimum; $s = q + 1$ at $N = 2q$ is included in H4. Thus, uniformly for $0 \leq t \leq 1$,

$$|\delta_{r,N}(t) - \delta_{r,N}(0)| \leq \log(1 + t\alpha_N) \leq O_h(q) = o(kq). \quad (\text{IQR23})$$

This proves the original-kernel coefficient transfer through this actual scalar adjacent source. For the activated two-column ACC24 update the exact counterpart is $\sum_{j=1}^{\min(m,r,2)} \log(1 + \lambda_j(U^*G_N(\eta)U))$. It controls the change from that actual activated baseline and keeps both column cross terms; it does not evaluate the initial angle.

The exact kernel-cost identity, now with a finite angle bound, is

$$\overline{F}_K(t) = F_K(t) + \delta(t) - \delta(0). \quad (\text{IQR19})$$

It follows by subtracting $\log \det \hat{H}_K(t) = \log \det H_K(t) - \delta(t)$ at the two ends of the same path. For a varying period, the original π_r also varies and the full derivative remains

$$\partial(\pi_r G^{-1} \pi_r^*) = (\partial \pi_r) G^{-1} \pi_r^* + \pi_r G^{-1} (\partial \pi_r)^* - \pi_r G^{-1} (\partial G) G^{-1} \pi_r^*.$$

IQR15-19 concerns the fixed-period source path, not deletion of these two period-frame terms.

5 The proper-source minimum and exact finite current receiver

For the proper-source branch retain NP30-31 [4] in its proved coprime, two-active period domain. Its target minimum is

$$H_e^p = F_{R,e}^* \mathcal{D}_e^p F_{R,e} - F_{R,e}^* \mathcal{D}_e^p J_* (J_*^* \mathcal{D}_e^p J_*)^{-1} J_*^* \mathcal{D}_e^p F_{R,e}.$$

Both H_e^p and its independent source metric Q_e^p are positive on that domain. With their common coefficient rank $r_p = O(k)$, use the ordered inverse-power comparison frame at the four separate reference cutoffs $\hat{N}_e = q - 1, q, 2q - 1, 2q$. The maps $U_{r_p} d \mapsto d$ and $\Lambda U_{r_p} d \mapsto [F_{R,e} d]$ into the complete proper target quotient are well defined: U_{r_p} and ΛU_{r_p} are injective by IVO. They form the specified commuting square. Taking determinants in these same coefficient coordinates proves

$$\begin{aligned} \mathcal{M}^p &= \mathcal{R}_e \left[\log \frac{\det H_e^p}{\det H_{r_p, \hat{N}_e}^B} - \log \frac{\det Q_e^p}{\det H_{r_p, \hat{N}_e}^E} \right] - \mathcal{R} \delta_{r_p, N} \\ &= \mathcal{R}_e \left[\log \frac{\det H_e^p}{\det H_{r_p, \hat{N}_e}^B} - \log \frac{\det Q_e^p}{\det H_{r_p, \hat{N}_e}^E} \right] + o(kq). \end{aligned} \quad (\text{IQR20})$$

The shared evaluated coefficient cancels. The two comparison determinants still contain the independent source and its full target minimum, and are not assigned values by that cancellation.

Finally let $x = U_r d$, $\Gamma = I_K^* G_N U_r$ and $\kappa = H_K^{-1} \Gamma d$. In the exact orthogonal decomposition $E = K \oplus P_B V_r \oplus (K + V_r)^\perp$ its coordinates are $(\kappa, d, 0)$ and the first two metric blocks are H_K, H_r^B . For the unchanged arithmetic action A , direct pairing proves

$$z = \kappa^* H_K A_{K,W} d, \quad w = d^* H_r^B A_{W,K} \kappa. \quad (\text{IQR21})$$

This computes both original cross currents in dimension $m + r$. The product $\text{Re}(\bar{z}w)$ retains all phases; no sign is supplied without evaluating these actual blocks. For a second action one must also keep $P_{K+V_r}AP_{\text{rem}}AP_{K+V_r}$, as follows by inserting $I = P_{K+V_r} + P_{\text{rem}}$ between the two copies of A . Thus the smaller receiver has an exact full-action relation.

$$\begin{array}{ccc}
E, G_N & \xrightarrow{\Lambda} & B, Q_{B,N} \\
\pi_r \downarrow & & \downarrow \rho_r \\
E/V_r, \bar{G}_{r,N} & \xrightarrow{\bar{\Lambda}_r} & B/\Lambda V_r, \bar{Q}_{r,N} \\
M^r \downarrow & & \\
E/\mathcal{P}_{<r} \text{ at cutoff } N+r & & \text{kernel metric: IQR3-6} \\
& & \text{complete source relations retained}
\end{array}$$

Figure 1: Exact quotient and shifted-source maps. The horizontal map's degree-zero cohomology is the original kernel. The vertical source map removes V_r ; multiplication moves its target to the low polynomial quotient with the displayed changed cutoff. Every norm is the attained original norm, not an unweighted coefficient norm.

$$\begin{array}{c}
\text{Tr}(P_{\text{rem}}e^{-\tau M^\dagger M}) \\
\leq \eta_N^2 e^{-\tau s_N^2} + (q-1)e^{-\tau/D_N^2} \\
\leftarrow A_{K,W}, A_{W,K} \rightarrow \\
\begin{array}{|c|c|c|}
\hline
K & P_B V_r & (K + V_r)^\perp \\
H_K & H_r^B & \bar{Q}_r \\
\hline
\end{array} \\
\leftarrow \text{complete higher-action return} \rightarrow
\end{array}$$

Figure 2: Exact orthogonal metric blocks, drawn schematically rather than proportional to dimension. IQR8-10 bounds the residual heat; IQR21 keeps both complex current cross blocks and the return through the third block. A positive heat estimate is not used to assign a complex-current sign.

References

- [1] Split-Zero research programme, The original observed inverse-power flag and its relative late heat, IFH1-21, especially IFH10-21.
- [2] Independent finite derivation, IVO1-26, complete finite constants.
- [3] Split-Zero research programme, The exact closed equilibrium coefficient, ECL1-17. Its full proof is retained in the accompanying cumulative source.
- [4] Split-Zero research programme, NP18-19, and NP30-31, full proper-source minimum.
- [5] Split-Zero research programme, NG2-3, full original source comparison and multiplication.

- [6] Supplied mathematical continuation, S1-45, especially S12-16 and S39-45.
- [7] Supplied mathematical continuation, H1-9, full original outgoing profile.
- [8] Supplied mathematical continuation, Original-metric heat transitions and complete mixed-family control, H17-23, and C6-10. The complete original text and TeX accompany this edition.
- [9] The Stacks Project Authors, Tag 0115, acyclic complexes and quasi-isomorphisms; original author source cochain homotopies and the definition. The finite contraction is supplied explicitly in received equations (15)-(18); the terminology is not a new general construction.
- [10] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw, with W. P. Reinhardt, NIST DLMF Chapter 18, Meixner-Pollaczek recurrence and generating function. The original Gamma mass and recurrence in S12-16 retain these human-source conventions. Their complete specialized estimates are proved in the cited programme sources.

Part II

Independent finite quotient and contraction proofs

6 Independent finite derivation of the inverse-quotient identities

6.1 Scope, inputs and reading record

This note verifies equations (3)–(31) of the user-supplied manuscript *The inverse-power theorem gives a quantitative reduction of the original observation complex*. It is an independent finite-dimensional derivation, not a claim to have re-proved the arithmetic inverse-power estimate, the Gamma division estimate, the monic-profile limit, or a Riemann-hypothesis endpoint.

The supplied text is at SUPPLIED_INVERSE_SECTOR_REDUCTION.md. Its equations (3)–(31) were read in full. The preceding programme manuscript *The original observed inverse-power flag and its relative late heat*, INVERSE_FLAG_AND_RELATIVE_HEAT.tex, was read through IFH10 to check the identity of the imported finite comparison. Its public source is the complete IFH proof. The exact source SHA-256 is 468c92ee4224a3659be44f122efb9fbfc0c2a469624ff5da6d16280aa069a6ca. In particular IFH1 defines the original observation metric, and IFH2–IFH7 give the finite source/observation sandwich and its passage to attained subquotients. This is programme mathematics, not an original human primary paper. No external primary-source theorem is silently attributed to it.

The mathematical task received from the root was to derive the Schur, quotient, chain-homotopy, inverse-division and observed-feedback identities, preserving the original objects and identifying any exact correction with a proof. Root retains source tracing and all later equations. No owner source, remote repository, or publication was changed by this task.

6.2 1. Original finite objects and the minimum formula

All adjoints denoted by a star in coordinate formulas are conjugate transposes. A metric matrix records a Hermitian form, not a replacement of the original form by the identity.

Let the full original polynomial be monic of degree q with $Q(0)$ nonzero. The algebra is $E = C[y]/(Q)$, represented in the complete ordered basis $1, y, \dots, y^{q-1}$. Multiplication by y is the invertible operator M . Let G be the specified positive Hermitian metric on E , let $\Lambda: E \rightarrow B$ be onto, and let $I_K: C^m \rightarrow E$ be the fixed full-column frame of $K = \ker \Lambda$. Set

$$H_K = I_K^* G I_K, \quad Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L_B = G^{-1} \Lambda^* Q_B.$$

For every onto matrix A from a space with metric H , the unique least-norm solution of $Ax=b$ is

$$S_A b = H^{-1} A^* (A H^{-1} A^*)^{-1} b.$$

Indeed $A S_A = I$, and for $z \in \ker A$, $z^* H S_A b = z^* A^* (A H^{-1} A^*)^{-1} b = 0$. Every solution is $S_A b + z$; its squared norm is $b^* (A H^{-1} A^*)^{-1} b + z^* H z$. This proves attainment,

uniqueness, the quotient metric, and the fact that S_A is an isometry from that quotient metric. It also proves A is a contraction and that S_{AA} is the original-metric orthogonal projection onto $(\ker A)^\perp$.

Apply this to Λ . Then $P_B = L_B \Lambda$ is the original G -orthogonal projection onto $K^\perp$. The complementary projection is

$$P_K = I_K H_K^{-1} I_K^* G, \quad P_B + P_K = I_E.$$

For $0 \leq r \leq q$, retain the ordered inverse-power map

$$U_r d = \sum_{j=1}^r d_j [y^{-j}], \quad V_r = \text{im } U_r.$$

It is injective: multiplying $U_r d = 0$ by M^r yields $\sum_{j=1}^r d_j y^{r-j} = 0$ in E , and the representative has degree less than q . Thus all its coefficients vanish. In the observation sector under consideration the imported IP lower factor is positive, so $\Phi_r = \Lambda U_r$ is also injective. Equivalently $K \cap V_r = 0$. Define

$$H^E = U_r^* G U_r, \quad H^B = \Phi_r^* Q_B \Phi_r, \quad \Gamma = I_K^* G U_r.$$

The finite input used below is precisely $a H^E \preceq H^B \preceq H^E$, with $a > 0$. Because H^E is positive, this also entails $a \leq 1$. The stated arithmetic IP theorem supplies a and its lower bound $\exp(-\varepsilon)$; this finite derivation does not change that input.

6.3 2. The two Schur complements, their eigenvalues and their exact loss

Since P_B is G -self-adjoint and idempotent,

$$H^B = U_r^* G P_B U_r = H^E - U_r^* G P_K U_r = H^E - \Gamma^* H_K^{-1} \Gamma. \quad (\text{F3})$$

For $z \in \mathbb{C}^m$ the quotient norm of $I_K z$ modulo V_r is the attained minimum of $(I_K z + U_r d)^* G (I_K z + U_r d)$ over all d . Completing the exact quadratic form yields the minimizing coefficient $d = -(H^E)^{-1} \Gamma^* z$ and hence

$$\hat{H}_K = H_K - \Gamma (H^E)^{-1} \Gamma^*. \quad (\text{F4})$$

This minimum remains positive because $K \cap V_r = 0$. The minimum representative is $(I - P_V) I_K z$, where $P_V = U_r (H^E)^{-1} U_r^* G$ is the original orthogonal projection onto V_r .

For the finite spectral comparison define, without altering either physical metric,

$$T = H_K^{-1/2} \Gamma (H^E)^{-1/2}.$$

The original metric matrices factor exactly as

$$H^B = (H^E)^{1/2} (I_r - T^* T) (H^E)^{1/2}, \quad \hat{H}_K = H_K^{1/2} (I_m - T T^*) H_K^{1/2}.$$

The positive eigenvalues of $T^* T$ and $T T^*$ coincide, with multiplicity: if $T^* T v = \mu v$ and $\mu > 0$, then $T v \neq 0$ and $T T^* (T v) = \mu T v$; the inverse correspondence on this eigenspace is $\mu^{-1} T^*$. Their number is $\nu = \text{rank } T = \text{rank } \Gamma$. The remaining eigenvalues are zero. IP gives $0 \preceq T^* T \preceq (1-a) I_r$, so the same nonzero-eigenvalue calculation gives $0 \preceq T T^* \preceq (1-a) I_m$. Consequently

$$a H_K \preceq \hat{H}_K \preceq H_K. \quad (\text{F5})$$

One can obtain the determinant equality without a spectral argument by taking the determinant of the same block Gram $\begin{pmatrix} H_K & \Gamma \\ \Gamma^* & H^E \end{pmatrix}$ in its two block orders. The equality is

$$\det H_K \det H^B = \det H^E \det \hat{H}_K, \quad \delta := \log \frac{\det H^E}{\det H^B} = \log \frac{\det H_K}{\det \hat{H}_K} = -\log \det(I_r - T^*T). \quad (\text{F6})$$

If the ν nonzero eigenvalues of T^*T are μ_j , then $\delta = -\sum_{j=1}^{\nu} \log(1 - \mu_j)$. Every factor $1 - \mu_j$ lies in $[a, 1]$. Thus

$$0 \leq \delta \leq \nu \log(1/a) \leq \min(m, r)\varepsilon \quad \text{when } a \geq e^{-\varepsilon}. \quad (\text{F7})$$

This proves the precise rank loss in (7), not a loss counted r times when $r > m$. With the manuscript's stated uniform $\varepsilon = o(q)$ and $m = O(k)$, it implies $\delta = o(kq)$ exactly as in (8). This is a propagation of the stated input, not a separate verification of its arithmetic range.

For the four-cutoff signs $+, +, -, -$, all four losses lie in the same interval $[0, d]$, where $d = \min(m, r)\varepsilon$. Their signed sum belongs to $[-2d, 2d]$: the sum of the two positively signed losses and the sum of the two negatively signed losses each belong to $[0, 2d]$. Since $\log \det \hat{H} = \log \det H_K - \delta$, this proves the factor two in (9). No cancellation beyond the common nonnegative interval has been assumed.

6.4 3. Explicit quotient coordinates and their attained metric

Let rem_Q denote the full degree-below- q representative. Define

$$\pi_r[p] = \mathcal{D}^r(\text{rem}_Q(y^r p)), \quad \mathcal{D}^r f = (f - \text{Tay}_{<r} f)/y^r. \quad (\text{F10})$$

This is well-defined on E , because replacing p by $p + Qa$ does not change $\text{rem}_Q(y^r p)$. It maps into $P_{<q-r}$. Its kernel consists exactly of classes whose $M^{\wedge r}$ -image is in $P_{<r}$. Since $M^{\wedge r} V_r = P_{<r}$, invertibility of $M^{\wedge r}$ gives $\ker \pi_r = V_r$. Let ι_r be the ordinary degree-below- $(q-r)$ inclusion. For p in this coefficient space, $y^r p$ has degree below q and its first r coefficients vanish; hence $\pi_r \iota_r p = p$. Thus π_r is onto.

The general minimum calculation in Section 1 yields

$$\overline{G}_r = (\pi_r G^{-1} \pi_r^*)^{-1}, \quad S_r = G^{-1} \pi_r^* \overline{G}_r. \quad (\text{F11})$$

Here S_r is the minimum section, and $S_r \pi_r = I - P_V$. The quotient coordinate metric is therefore the attained one, not the raw polynomial Gram. Direct substitution gives

$$(\pi_r I_K)^* \overline{G}_r (\pi_r I_K) = I_K^* G (I - P_V) I_K = \hat{H}_K.$$

The physical substitution $S = c + iy$ is retained by its full triangular map

$$(T_c)_{ab} = \binom{b}{a} c^{b-a} i^a \quad (a \leq b), \quad (T_c)_{ab} = 0 \quad (a > b).$$

It sends S -coefficients to y -coefficients. Therefore $I_K^y = T_c I_K^S$ and $G^S = T_c^* G^y T_c$, and both sides give the same fixed-kernel Gram. The diagonal phases i^a are retained.

6.5 4. The quotient observation, its metric and the chain contraction

Because Φ_r is injective, choose r rows I with invertible minor Φ_I , and let J be the remaining row indices. With coordinate selectors R_I and R_J , define the fixed algebraic quotient matrix

$$\rho_r = R_J - \Phi_J \Phi_I^{-1} R_I.$$

It kills Φ_r . Its restriction to the coordinate complement supported on J is the identity, so it is onto and has kernel $W_r = \text{im } \Phi_r$. This formula depends on the original algebraic maps, not on the cutoff metric. Define

$$\bar{\Lambda}_r = \rho_r \Lambda \iota_r.$$

Since $x - \iota_r \pi_r x$ lies in V_r and $\rho_r \Lambda$ kills V_r , $\text{bar } \Lambda_r \pi_r = \rho_r \Lambda$. Onto-ness follows from that of $\rho_r \Lambda$. If $\text{bar } \Lambda_r \pi_r x = 0$, then $\Lambda x = \Lambda v$ for some $v \in V_r$, so $x - v \in K$ and $\pi_r x \in \pi_r K$. Conversely $\pi_r K$ is killed. The restriction $\pi_r : K \rightarrow \ker \text{bar } \Lambda_r$ is injective and onto. This proves all arrows and exactness in (12)-(13).

For the target quotient the attained metric is

$$\bar{Q}_r = (\rho_r Q_B^{-1} \rho_r^*)^{-1}.$$

The inverse of this matrix is exactly

$$\begin{aligned} \rho_r \Lambda G^{-1} \Lambda^* \rho_r^* &= \bar{\Lambda}_r \pi_r G^{-1} \pi_r^* \bar{\Lambda}_r^* \\ &= \bar{\Lambda}_r \bar{G}_r^{-1} \bar{\Lambda}_r^*. \end{aligned}$$

Inverting proves (14) and equality of the two orders of complete minimum, without identifying either quotient metric with a raw coefficient Gram.

Now set

$$h = U_r (H^B)^{-1} \Phi_r^* Q_B : B \longrightarrow E. \quad (\text{F15})$$

Then $h \Phi_r = U_r$, so $h \Lambda$ is idempotent with image V_r . Its kernel is the preimage under $\bar{\Lambda}$ of the Q_B -orthogonal complement to W_r . Meanwhile $P_W = \Lambda h = \Phi_r (H^B)^{-1} \Phi_r^* Q_B$ is the original Q_B -orthogonal projection onto W_r . The degree-zero projection $h \Lambda$ need not be orthogonal.

Write $hb = U_r d$. The equality $d = (H^B)^{-1} \Phi_r^* Q_B b$ gives

$$\|hb\|_G^2 = d^* H^B d \leq a^{-1} d^* H^B d = a^{-1} \|P_W b\|_{Q_B}^2 \leq a^{-1} \|b\|_{Q_B}^2.$$

Thus $\|h\| \leq a^{-1/2}$, proving (16) with its exact metric domains.

Let $S_\rho = Q_B^{-1} \rho_r^* \bar{Q}_r$ be the minimum section of ρ_r . Since $S_\rho \rho_r = I - P_W$, the fully explicit chain section of $p = (\pi_r, \rho_r)$ is

$$j^0 = (I_E - h \Lambda) S_r, \quad j^1 = S_\rho. \quad (\text{F17a})$$

First $\pi_r j^0 = \pi_r S_r = I$ and $\rho_r j^1 = I$. Secondly,

$$\Lambda j^0 = (I_B - \Lambda h) \Lambda S_r = S_\rho \rho_r \Lambda S_r = S_\rho \bar{\Lambda}_r = j^1 \bar{\Lambda}_r.$$

Thus j is a chain map. The difference $x - S_r \pi_r x$ belongs to V_r and $I - h \Lambda$ kills V_r ; hence $j^0 \pi_r = I - h \Lambda$. The other degree gives $j^1 \rho_r = I - \Lambda h$. Regard $h : B \rightarrow E$ as a degree -1 map, with its other component zero. For the differential $d = \Lambda$ these two equalities are exactly

$$I - jp = dh + hd. \quad (\text{F17b})$$

This proves the finite chain-homotopy equivalence (18). It fixes K pointwise: for $x \in K$, $j^0 \pi_r x = (I - h\Lambda)x = x$. No infinite theta complex is introduced by this calculation.

6.5.1 Complete polynomial relation spaces remain present

Let the actual source P_N have its positive source metric, and let $J_N: P_N \rightarrow E$ be the full remainder map, with minimum section L_N . Then $J_N L_N = I$ and L_N is an isometry from G . The lifted homotopy $\tilde{h} = L_N h$ has exactly the same norm. Its range is $L_N V_r$. Moreover $\tilde{h} \Lambda J_N$ vanishes on the complete original relation space $\ker J_N = QP_{N-q}$, and this relation space is orthogonal to $L_N E$ and therefore to $L_N V_r$.

There is a corresponding reduction of $[P_N \rightarrow B]$ by the contractible subcomplex $[L_N V_r \rightarrow W_r]$. Its source quotient is $P_N / (L_N V_r)$, not an identification of P_N with E : every original Q -relation remains in degree-zero cohomology. The same proof with differential ΛJ_N and homotopy $L_N h$ gives the contraction. This specifies exactly what the manuscript's polynomial-source lift preserves.

6.6 5. Exact determinant allocation and its fixed frame factor

The Gram of the augmented kernel $K + V_r$ in the fixed ordered frame $[I_K, U_r]$ has blocks $H_K, \Gamma, \Gamma^*, H^E$. Its determinant is

$$\det([I_K, U_r]^* G [I_K, U_r]) = \det H_K \det H^B. \quad (\text{F19})$$

Choose any fixed right inverse Z of ρ_r and set $C = [\Phi_r, Z]$. This square matrix is invertible. In that frame the Q_B Gram has leading block H^B and lower Schur complement $\bar{Q}_r$. Therefore the complete equality is

$$|\det C|^2 \det Q_B = \det H^B \det \bar{Q}_r,$$

or

$$\log \det \bar{Q}_r = \log \det Q_B - \log \det H^B + 2 \log |\det C|. \quad (\text{F20})$$

C is fixed across the four cutoffs, so the last term cancels under their signs. This is precisely (20), including the coefficient-frame constant before the return is taken. Combined with (F19), the signs of the two redistributed terms in (21) are $+$ and $-$. Any use of the evaluated coefficient C_{frac} belongs to the imported IFH coefficient theorem; the allocation itself has no asymptotic error.

6.7 6. Inverse multiplication as exact Taylor division between quotients

For $x \in E$ retain its actual minimum source representative $L_N x$, and write

$$L_N x = \sum_{j=0}^{s-1} c_j(x) y^j + y^s \mathcal{D}^s L_N x.$$

Apply J_N , and then multiply by $M^(-s)$. The result is

$$M^{-s}x = J_N \mathcal{D}^s L_N x + \sum_{n=1}^s c_{s-n}(x)[y^{-n}]. \quad (\text{F22})$$

Thus the ordering in $F_{-}(s, N)$ is exactly $(F_{-}(s, N)x)_n = c(s-n)(x)$, as in (22). No assertion has been made that polynomial Taylor coefficients themselves are well-defined on E ; the map L_N supplies their specific representative.

For the norm in (23), the supplied Gamma comparison has the form $\text{ell}_{-}(k, N)$ $H_{\text{Gamma}} \leq H_{\text{native}}$ on degrees $\leq N$ and $H_{\text{native}} \leq u_k H_{\text{Gamma}}$ on the output range. Apply these two comparisons on the outside of the s -fold division estimate, giving

$$\|\mathcal{D}^s f\|_{\text{native}} \leq \sqrt{u_k} \|\mathcal{D}^s f\|_{\Gamma} \leq \sqrt{u_k} (a_N^{\Gamma})^s \|f\|_{\Gamma} \leq \sqrt{u_k / \ell_{k,N}} (a_N^{\Gamma})^s \|f\|_{\text{native}}.$$

The source minimum L_N is an isometry and J_N is a contraction, so $\|J_N D^s L_N\| \leq B_{-}(s, N)$. Only one comparison ratio occurs. This derives the use of the imported division bound, not its original Gamma proof.

Since $M^{-s}V_r \subset V_{r+s}$, the map

$$\mathcal{I}_{r,s} = \pi_{r+s} M^{-s} \iota_r : E/V_r \rightarrow E/V_{r+s}$$

is well-defined for $r+s \leq q$. For the canonical coordinate p of degree below $q-r$, calculate

$$\pi_{r+s} M^{-s} \iota_r p = \mathcal{D}^{r+s} \text{rem}_Q(M^{r+s} M^{-s} p) = \mathcal{D}^{r+s}(y^r p) = \mathcal{D}^s p. \quad (\text{F24})$$

The second equality means the representative of the algebraic product: $M^{r+s} M^{-s} p = M^r p$, whose representative is $y^r p$ since its degree is below q . Hence the kernel is exactly $P_{<s}$ inside $P_{<q-r}$ and the map is onto $P_{<q-r-s}$. Equivalently, $M^s V_{r+s} = V_r \oplus \mathcal{P}_{<s}$: multiplying by M^r sends these two summands to $P_{<r}$ and $\text{span}(y^r, \dots, y^{r+s-1})$, disjoint complete monomial ranges of degree below q . This proves (25), including its first injection.

For the norm, take the actual minimum representative $x = S_{-r} z$ of a quotient class z . The last term of (F22) belongs to $V_{-s} \subset V_{-(r+s)}$, so

$$\mathcal{I}_{r,s} z = \pi_{r+s} J_N \mathcal{D}^s L_N S_r z.$$

Each outer quotient map is a contraction and S_r an isometry in the attained metrics. This gives (26). Directly from $M^{-t} M^{-s} = M^{-s-t}$, or from $\mathcal{D}^t \mathcal{D}^s = \mathcal{D}^{s+t}$, one obtains (27) with its degree-changing domain and codomain. No fixed quotient is being treated as M -invariant.

6.8 7. Complete observed leakage and composition memory

Let $\text{bar} L_r$ be the attained minimum section of $\text{bar} \Lambda_r$ and let $\text{bar} P^{\wedge} K_r = I - \text{bar} L_r \text{bar} \Lambda_r$. It is the original quotient-metric orthogonal projection onto $\ker \text{bar} \Lambda_r$. The observed map is

$$\mathcal{B}_{r,s} = \bar{\Lambda}_{r+s} \mathcal{I}_{r,s} \bar{L}_r.$$

The section is an isometry and $\text{bar} \Lambda_{(r+s)}$ is a contraction. Consequently $\|\text{bar} L_{(r,s)}\| \leq B_{-}(s, N)$, exactly (28). Insert $I = \text{bar} L_r \text{bar} \Lambda_r + \text{bar} P^{\wedge} K_r$ to obtain

$$\bar{\Lambda}_{r+s} \mathcal{I}_{r,s} = \mathcal{B}_{r,s} \bar{\Lambda}_r + \bar{\Lambda}_{r+s} \mathcal{I}_{r,s} \bar{P}_r^K. \quad (\text{F29})$$

All three factors in the last term have norm ≤ 1 , $B_-(s,N)$, ≤ 1 in their actual metrics, proving the same leakage bound.

For the composition use the exact identity (27) and subtract the observed product. The middle difference is $I - \bar{L}_{r+s}\bar{L}_{r+s} = \bar{P}_{r+s}^K$, so

$$\mathcal{B}_{r,s+t} - \mathcal{B}_{r+s,t}\mathcal{B}_{r,s} = \bar{L}_{r+s+t}\mathcal{I}_{r+s,t}\bar{P}_{r+s}^K\mathcal{I}_{r,s}\bar{L}_r. \quad (\text{F30})$$

The sign is positive. The norm is at most $B_-(t,N)B_-(s,N)$. On the retained injective conductor domain $\ker \text{bar}\Lambda_-(r+s) = \pi_-(r+s)K$ has dimension m , and the rank is therefore at most m . Even outside injectivity of a particular inverse sector, the kernel dimension is at most m because its kernel remains $\pi_-(r+s)K$; the formulas involving its quotient metric still use its actual target quotient dimension.

Finally $\rho_-(r+s)\Lambda U_s = 0$ since $\Lambda V_s \subset W_-(r+s)$. Apply $\rho_-(r+s)\Lambda$ to (F22) and restrict the input by I_K . This proves

$$\rho_{r+s}\Lambda M^{-s}I_K = \rho_{r+s}\Lambda J_N \mathcal{D}^s L_N I_K. \quad (\text{F31})$$

The frame I_K is an isometry from its retained coordinate metric H_K to its physical image, and $\rho_-(r+s)\Lambda$ is a contraction for the attained target metric. This proves the stated norm bound, without quotienting or changing the metric on the original K .

6.9 8. Exact finite verification and conclusion

The companion `verify_exact.py` uses SymPy exact rational complex arithmetic, the full polynomial $Q = y^6 + 2y^4 - y^3 + 3y + 2$, an eight-dimensional positive source Gram with nonzero imaginary off-diagonal entries, its original quotient Gram, and a fixed onto observation and fixed two-dimensional kernel frame. The source metric is constructed explicitly, then G and all sections are obtained by their stated attained-minimum formulas. No floating or identity-metric replacement is used.

The script checks $r=1,2,3$, full Schur complements, all quotient and chain maps, both homotopy identities, the exact frame determinant factor, inverse Taylor decomposition, and the observed feedback identity. It requires the observed composition defect to be nonzero, so success cannot result merely from accidentally selecting an invariant observed sector. The actual result count and script hash are in `EXACT_CHECKS.json`.

No correction to the finite identities (3)-(31) was found. The two details made explicit here are already consistent with the supplied manuscript: (i) the fixed frame factor $2 \log|\det C|$ in (F20) exists before the four-cutoff return; (ii) the polynomial-source contraction quotients by L_{NV_r} and retains its entire Q -relation kernel, rather than replacing that larger source complex by $[E \rightarrow B]$.

Part III

Independent covariance-angle derivation

7 Independent covariance-angle derivation

Date: 21 September 2026.

This file checks IQR15–19 and the algebra of IQR2 in `INVERSE_QUOTIENT_RECEIVERS.tex`. It proves the finite matrix statements independently, retains the original metric and the complete source covariance, and spells out the four-cutoff estimate. It does not change the mathematical period domain or assert a new analytic estimate for the source covariance.

The exact finite angle estimate is correct. A stronger estimate using the individual covariance eigenvalues is proved below. The coefficient identity is correct when the retained endpoint value $\psi(0) = \log(4/\pi)$ is included explicitly. No determinant factor is missing when the frame on $K + V$ is the stated concatenation.

7.1 Reading and derivation record

The source inspected for this derivation was `INVERSE_QUOTIENT_RECEIVERS.tex`, SHA-256 42664D36E612DEB49550850DCAA15EF2774B58682A07CDF7F707DF5F3D3F9BA2. The whole file was read; the independent calculations below concern IQR2 and IQR13–19, together with the exact Schur identity used in IQR3. The separate endpoint definition was checked in `../publication_003/NATIVE_SMALLEST_SINGULAR_PROOF.tex`, S2–S3, lines 31–41, SHA-256 C7DD360325DC475B82CABBC65F2DC01D5C67711B85951A1837CD3E7F52415BBB. That reading was bounded to its opening theorem and definitions, lines 1–58. Its public source locator is S2–S3 in the retained smallest-singular proof. This file is a new finite-dimensional derivation, not a transcription of that source.

The proof is complete for the stated finite hypotheses. The asymptotic application uses the already stated IQR1 size estimate and the explicitly displayed covariance-size hypotheses; those earlier analytic results are not independently rederived here.

7.2 1. Exact objects, frames, and covariance

Let $E = \mathbb{C}^n$ in its fixed coefficient coordinates. Let G_0 be a positive-definite Hermitian matrix, and let Ω be a positive-semidefinite Hermitian matrix. For every real $t \geq 0$, set

$$G(t) = (G_0^{-1} + t\Omega)^{-1}. \quad (\text{CA1})$$

Every inverse in CA1 exists: for $x \neq 0$, $x^*(G_0^{-1} + t\Omega)x \geq x^*G_0^{-1}x > 0$. Use the inner product $\langle x, y \rangle_t = x^*G(t)y$, with conjugate linearity in the first argument.

Keep fixed subspaces $K, V \subset E$, with $K \cap V = \{0\}$, dimensions m, r , and fixed full column frames $I_K : \mathbb{C}^m \rightarrow E$, $U : \mathbb{C}^r \rightarrow E$. Set $S_0 = K + V$, and use precisely the concatenated frame $I_{S_0} = [I_K, U]$. Thus $\dim S_0 = m + r \leq n$. For $X = K, V, S_0$, write

$$H_X(t) = I_X^* G(t) I_X, \quad \Gamma(t) = I_K^* G(t) U. \quad (\text{CA2})$$

Here $I_V = U$. All three Gram matrices are positive definite because their frames are injective. Define

$$\delta(t) = \log \det H_K(t) + \log \det H_V(t) - \log \det H_{S_0}(t). \quad (\text{CA3})$$

The logarithms are the real logarithms of positive determinants. If one subspace has dimension zero, its determinant is 1 and $\delta \equiv 0$; every assertion below then holds with a zero right side.

For the source in IQR13, the matrix in CA1 is the full matrix

$$\Omega = F_Z H_Z^+ F_Z^*, \quad H_Z \succeq 0, \quad \ker H_Z \subseteq \ker F_Z. \quad (\text{CA4})$$

Indeed $\Omega = BB^*$ with $B = F_Z(H_Z^+)^{1/2}$, so $\Omega \succeq 0$. If Π is the Euclidean projection onto $\text{ran } H_Z$, the kernel hypothesis gives $F_Z = F_Z \Pi$. The matrix $(H_Z^+)^{1/2}$ is invertible on $\text{ran } H_Z$, so $\text{rank } \Omega = \text{rank } B = \text{rank } F_Z$. Neither this factorization nor anything below removes a cross term of H_Z , H_Z^+ , or Ω .

7.3 2. The Schur identity and exact quotient meaning

In the stated concatenated frame,

$$H_{S_0}(t) = \begin{pmatrix} H_K(t) & \Gamma(t) \\ \Gamma(t)^* & H_V(t) \end{pmatrix}. \quad (\text{CA5})$$

Suppress t within this section. Left multiplication by $\begin{pmatrix} I & 0 \\ -\Gamma^* H_K^{-1} & I \end{pmatrix}$ is determinant preserving and gives an upper block triangular matrix. Consequently

$$\det H_{S_0} = \det H_K \det(H_V - \Gamma^* H_K^{-1} \Gamma). \quad (\text{CA6})$$

Eliminating the other off-diagonal block gives

$$\det H_{S_0} = \det H_V \det(H_K - \Gamma H_V^{-1} \Gamma^*). \quad (\text{CA7})$$

Both Schur complements are positive definite. For example, for $a \neq 0$, insert $(a, -H_V^{-1} \Gamma^* a) \neq 0$ into the positive quadratic form CA5 to obtain $a^*(H_K - \Gamma H_V^{-1} \Gamma^*)a > 0$.

Let $\pi : E \rightarrow E/V$ be represented by any fixed onto matrix whose kernel is V . Its attained quotient metric is

$$\overline{G}(t) = (\pi G(t)^{-1} \pi^*)^{-1}. \quad (\text{CA8})$$

To verify this formula directly, for a quotient coordinate b , put $x_* = G(t)^{-1} \pi^* \overline{G}(t)b$. Then $\pi x_* = b$, and for every $v \in \ker \pi$, $v^* G(t)x_* = v^* \pi^* \overline{G}(t)b = 0$. Every other lift equals $x_* + v$, whose squared norm is $x_*^* G(t)x_* + v^* G(t)v$. Hence x_* is the unique minimum lift and the attained squared norm is $b^* \overline{G}(t)b$.

For the quotient image of $I_K a$, all lifts have the form $I_K a + Uz$. Their squared norms equal

$$a^* H_K a + a^* \Gamma z + z^* \Gamma^* a + z^* H_V z.$$

Completion of this quadratic form gives the unique minimizer $z = -H_V^{-1} \Gamma^* a$. Therefore the quotient kernel Gram is exactly

$$\widehat{H}_K(t) = (\pi I_K)^* \overline{G}(t) (\pi I_K) = H_K(t) - \Gamma(t) H_V(t)^{-1} \Gamma(t)^*. \quad (\text{CA9})$$

The disjointness hypothesis makes πI_K injective. Equations CA3 and CA7 give

$$\delta(t) = \log \det H_K(t) - \log \det \widehat{H}_K(t). \quad (\text{CA10})$$

For the original observation $\Lambda : E \rightarrow B$ with $\ker \Lambda = K$, the identical minimum calculation gives $H_V^B(t) = H_V(t) - \Gamma(t)^* H_K(t)^{-1} \Gamma(t)$, using the attained metric $(\Lambda G(t)^{-1} \Lambda^*)^{-1}$ on B . Thus

$$\delta(t) = \log \det H_V(t) - \log \det H_V^B(t). \quad (\text{CA11})$$

This proves the exact morphism between the two angle losses. It retains the physical subspaces and all cross-Gram entries.

For an explicit expression, define

$$B_t = H_K(t)^{-1/2} \Gamma(t) H_V(t)^{-1/2}.$$

From CA6,

$$\delta(t) = -\log \det(I_r - B_t^* B_t) = - \sum_{j=1}^{\min(m,r)} \log(1 - c_j(t)^2) \geq 0, \quad (\text{CA12})$$

where the $c_j(t)$ are the singular values of B_t , including zeros. Positivity of the Schur complement proves $c_j(t) < 1$.

There is no unstated frame constant in these equations. Replacing I_K, U by $I_K T_K, U T_V$, with both matrices invertible, changes the three log determinants in CA3 by $2 \log |\det T_K|$, $2 \log |\det T_V|$, and their sum. These changes cancel. If instead a separate frame $I_{S_0} T$ is used on S_0 while the first two frames are retained, CA3 acquires the constant $-2 \log |\det T|$. Its nonnegativity need not survive, although the change $\delta(t) - \delta(0)$ is unchanged for fixed T . This is why the concatenation in IQR15 matters for IQR18.

7.4 3. Exact differentiated identity

Differentiating $(G_0^{-1} + t\Omega)G(t) = I$ gives

$$G'(t) = -G(t)\Omega G(t). \quad (\text{CA13})$$

For a fixed full frame I_X , the determinant differential follows from the cofactor identity for the determinant:

$$\frac{d}{dt} \log \det H_X(t) = \text{Tr}(H_X(t)^{-1} I_X^* G'(t) I_X).$$

Define the original-metric projection and its exact Euclidean transform by

$$P_X(t) = I_X H_X(t)^{-1} I_X^* G(t), \quad \tilde{P}_X(t) = G(t)^{1/2} P_X(t) G(t)^{-1/2}. \quad (\text{CA14})$$

The first operator is idempotent, fixes X , and satisfies $G(t)P_X(t) = P_X(t)^* G(t)$. The second is $G(t)^{1/2} I_X H_X(t)^{-1} I_X^* G(t)^{1/2}$, so it is the Euclidean orthogonal projection onto the actual transformed subspace $G(t)^{1/2} X$.

Substitution of CA13 and cyclicity of the finite trace give

$$\frac{d}{dt} \log \det H_X(t) = -\text{Tr}(G(t)^{1/2} \Omega G(t)^{1/2} \tilde{P}_X(t)). \quad (\text{CA15})$$

Consequently, for $t > 0$,

$$t\delta'(t) = -\text{Tr}(D_t A_t), \quad D_t = tG(t)^{1/2} \Omega G(t)^{1/2}, \quad A_t = \tilde{P}_K(t) + \tilde{P}_V(t) - \tilde{P}_{S_0}(t). \quad (\text{CA16})$$

This proves IQR15. It differentiates the original fixed-frame Gram matrices. No derivative of a moving transformed frame has been omitted: the transformed projections occur only after evaluating that derivative by CA15.

7.5 4. Complete spectrum of the projection-pair operator

Fix t , and write $\mathcal{K} = G(t)^{1/2}K$, $\mathcal{V} = G(t)^{1/2}V$. Their intersection remains zero because $G(t)^{1/2}$ is invertible. The matrices

$$T_K = G(t)^{1/2}I_K H_K(t)^{-1/2}, \quad T_V = G(t)^{1/2}U H_V(t)^{-1/2}$$

have orthonormal columns, and $T_K^* T_V = B_t$. Choose a singular-value decomposition of B_t , rotating these two orthonormal frames accordingly. If $p = \min(m, r)$, the resulting unit vectors satisfy

$$e_i^* e_j = \delta_{ij}, \quad f_i^* f_j = \delta_{ij}, \quad e_i^* f_j = c_i \delta_{ij}, \quad (\text{CA17})$$

for paired indices, with all additional cross products zero. Here $e_i \in \mathcal{K}$ and $f_i \in \mathcal{V}$, not the untransformed physical subspaces. By Cauchy-Schwarz, $c_i \leq 1$. Equality would give $f_i = e_i$, contradicting $\mathcal{K} \cap \mathcal{V} = 0$. Thus $0 \leq c_i < 1$, agreeing with CA12.

For each paired index put $h_i = (f_i - c_i e_i)/\sqrt{1 - c_i^2}$. Equations CA17 show that all pairs (e_i, h_i) are orthonormal and mutually orthogonal. The extra unpaired vectors are orthogonal to their sum. On the pair (e_i, h_i) , the two projections are

$$\tilde{P}_K = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \tilde{P}_V = \begin{pmatrix} c_i^2 & c_i \sqrt{1 - c_i^2} \\ c_i \sqrt{1 - c_i^2} & 1 - c_i^2 \end{pmatrix}.$$

The projection onto $\mathcal{K} + \mathcal{V}$ is the identity on that pair. Hence

$$A_t|_{\text{span}(e_i, h_i)} = \begin{pmatrix} c_i^2 & c_i \sqrt{1 - c_i^2} \\ c_i \sqrt{1 - c_i^2} & -c_i^2 \end{pmatrix}, \quad \det(\lambda I - A_t) = \lambda^2 - c_i^2. \quad (\text{CA18})$$

The eigenvalues on this pair are $+c_i, -c_i$. If $c_i = 0$, the entire block is zero. An unpaired vector lies in one subspace and is orthogonal to the other, so A_t is zero on it as well. Outside $\mathcal{K} + \mathcal{V}$, all three projections vanish. This accounts for every dimension of E .

Thus the nonzero eigenvalues of A_t are exactly $\nu(t)$ pairs $+c_i, -c_i$, where

$$\nu(t) = \text{rank } B_t = \text{rank}(I_K^* G(t) U) \leq \min(m, r). \quad (\text{CA19})$$

In particular, if $A_t = A_{t,+} - A_{t,-}$ is its positive/negative spectral decomposition, then

$$0 \preceq A_{t,+}, A_{t,-} \preceq I, \quad \text{rank } A_{t,+} = \text{rank } A_{t,-} = \nu(t). \quad (\text{CA20})$$

This proves the complete IQR16 spectrum, including the orthogonal and exterior directions.

7.6 5. Covariance spectrum and the sharp rank-dependent trace estimate

Set

$$C = G_0^{1/2} \Omega G_0^{1/2}, \quad d = \text{rank } \Omega = \text{rank } C, \quad c = \|C\|. \quad (\text{CA21})$$

Write the eigenvalues of C in decreasing order as $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. Direct multiplication proves

$$G(t) = G_0^{1/2}(I + tC)^{-1}G_0^{1/2}, \quad G_0^{-1/2}G(t)\Omega G_0^{1/2} = (I + tC)^{-1}C. \quad (\text{CA22})$$

Also $G(t)^{-1/2}(tG(t)\Omega)G(t)^{1/2} = D_t$. Therefore the positive matrix D_t has the exact ordered eigenvalues

$$\mu_j(t) = \frac{t\lambda_j}{1 + t\lambda_j}, \quad \text{rank } D_t = d \ (t > 0), \quad \|D_t\| = \frac{tc}{1 + tc}. \quad (\text{CA23})$$

The following elementary trace bound also records why a factor 2 is unnecessary. Let $D \succeq 0$, and let $0 \preceq B \preceq I$ have rank at most s . In an eigenbasis u_j of D , with descending eigenvalues μ_j , set $b_j = u_j^* B u_j$. Then $0 \leq b_j \leq 1$, and $\sum b_j = \text{Tr } B \leq s$. The decreasing order gives

$$\sum_j \mu_j b_j \leq \sum_{j=1}^s \mu_j. \quad (\text{CA24})$$

For completeness, when $1 \leq s < n$, the tail satisfies $\sum_{j>s} \mu_j b_j \leq \mu_s \sum_{j>s} b_j$, while $\sum_{j \leq s} \mu_j (1 - b_j) \geq \mu_s \sum_{j \leq s} (1 - b_j)$. The first total weight is at most the second because $\sum b_j \leq s$, proving CA24. The cases $s = 0$ and $s = n$ are immediate.

Apply CA24 separately to $A_{t,+}$ and $A_{t,-}$. Each of $a = \text{Tr}(D_t A_{t,+})$ and $b = \text{Tr}(D_t A_{t,-})$ is nonnegative and at most $\sum_{j=1}^{\nu(t)} \mu_j(t)$. Since $|a - b| \leq \max(a, b)$, not merely $a + b$, CA16 yields

$$|t\delta'(t)| \leq \sum_{j=1}^{\min(\nu(t), d)} \frac{t\lambda_j}{1 + t\lambda_j} \leq \ell \frac{tc}{1 + tc}, \quad \ell = \min(m, r, d). \quad (\text{CA25})$$

For $t > 0$, division by t gives an integrable bound down to zero. All original Grams are smooth and positive at zero, so δ is continuously differentiable there. Integration proves the stronger estimate

$$|\delta(t) - \delta(0)| \leq \sum_{j=1}^{\ell} \log(1 + t\lambda_j(C)) \leq \min(m, r, \text{rank } \Omega) \log(1 + t\|C\|).$$

(\text{CA26})

If $d = 0$, positive semidefiniteness gives $\Omega = 0$, and CA1 is constant. Thus CA26 also covers that case without dividing by a positive eigenvalue. The second inequality in CA26 is exactly IQR17.

The instantaneous cross-Gram rank $\nu(t)$ is available in CA25. It cannot in general be replaced throughout the integrated estimate by $\nu(0)$, because covariance activation may create nonzero cross-Gram entries. An exact example follows below.

7.7 6. Four original cutoffs and the original coefficient scale

Return to the original objects and cutoffs in the receiver:

$$q = (k + 1)^2, \quad k \equiv 1 \pmod{4}, \quad N = q - 1, q, 2q - 1, 2q, \quad m = 8k - 16, \quad 1 \leq r < q.$$

Keep every period, source, and finite positivity guard of IQR1 and IQR3. For each original cutoff use its actual $G_N(0)$, Ω_N , $C_N = G_N(0)^{1/2}\Omega_N G_N(0)^{1/2}$, and activation $t_N \geq 0$. The fixed-period argument applies separately at each cutoff; it does not identify these four matrices with one another. For the absolute-angle application

below, $G_N(0)$ is the canonical original metric unless the same initial-angle estimate has been proved for the actual different baseline.

For that canonical original metric, the previously proved base estimate is

$$0 \leq \delta_{r,N}(0) \leq \min(m, r)\varepsilon_{k,r}. \quad (\text{CA27})$$

The finite change estimate CA26 holds for every positive baseline, but it does not by itself prove CA27 at an already activated baseline. In particular, if an ACC path starts from a previously activated metric, its initial angle is the angle of that metric. The canonical IVO bound cannot be substituted for it without a proof connecting the two actual baselines.

Adding CA26 and using CA12 gives the finite IQR18 bound

$$0 \leq \delta_{r,N}(t_N) \leq L_N := \min(m, r)\varepsilon_{k,r} + \ell_N \log(1 + t_N \|C_N\|), \quad \ell_N = \min(m, r, \text{rank } \Omega_N). \quad (\text{CA28})$$

Let the original four-sign operator be $\mathcal{R}a = a_{q-1} + a_q - a_{2q-1} - a_{2q}$. For four quantities $0 \leq a_N \leq L_N$, set $P = a_{q-1} + a_q$, $M = a_{2q-1} + a_{2q}$. Then $0 \leq P \leq L_{q-1} + L_q$, $0 \leq M \leq L_{2q-1} + L_{2q}$, so

$$|\mathcal{R}\delta_{r,N}(t_N)| \leq \max\{L_{q-1} + L_q, L_{2q-1} + L_{2q}\}. \quad (\text{CA29})$$

This uses the signs and nonnegativity exactly; it is stronger than summing all four L_N .

Now suppose the original asymptotic regime is $r \log(q+2) = o(q)$, and the fixed source constants in IQR1 give

$$\varepsilon_{k,r} = O_{h,A}((k+Jr) \log(q+Jr+2)) = o(q). \quad (\text{CA30})$$

The last equality also follows directly when J is fixed: because $r < q$, the logarithm is $O(\log(q+2))$; because $q = (k+1)^2$, $k \log(q+2) = o(q)$; and the remaining term is small by the stated regime. Since $m = O(k)$,

$$\min(m, r)\varepsilon_{k,r} \leq m\varepsilon_{k,r} = o(kq). \quad (\text{CA31})$$

This deduction covers $r \leq k$ and $r > k$ without introducing a second regime or dropping an original coordinate.

For the covariance term require its exact activated size bound

$$\max_N \log(1 + t_N \|C_N\|) = O_h(q \log(q+2)), \quad \max_N \text{rank } \Omega_N \leq d_k, \quad d_k \log(q+2) = o(k). \quad (\text{CA32})$$

The first bound follows, for example, from the stated unactivated logarithmic-size bound and $0 \leq t_N \leq 1$, as in the nonnegative-parameter activation scales in IQR14. If activations larger than 1 are allowed, it is the activated expression in CA32 that must satisfy the displayed bound. Then

$$\ell_N \log(1 + t_N \|C_N\|) \leq d_k O_h(q \log(q+2)) = o(kq) \quad (\text{CA33})$$

uniformly over the four cutoffs. Equations CA29–33 prove

$$\mathcal{R}\delta_{r,N}(t_N) = o(kq). \quad (\text{CA34})$$

For every actual one-column or two-column source, CA4 gives $\text{rank } \Omega_N \leq 1$ or ≤ 2 , including dependent columns and complete off-diagonal source Grams. Here

$d_k \leq 2$, and $d_k \log(q+2)/k \rightarrow 0$. Thus, when CA27 and the actual-baseline size bound CA32 hold, CA34 applies at both ranks. If CA27 has not been proved at an already activated baseline, rank one or two still gives the angle-change and cost bounds in CA26 and CA37 under CA32; it does not alone give the absolute-angle conclusion CA34. The covariance's norm itself can grow exponentially on the allowed scale; the proof does not require the source change to be small. A general rank- q covariance does not satisfy the rank hypothesis in CA32, and no $o(kq)$ conclusion for it follows from this estimate alone.

7.8 7. The exact kernel-cost identity and its four-sign version

Use the determinant-loss convention

$$F_K(t) = \log \det H_K(0) - \log \det H_K(t), \quad \overline{F}_K(t) = \log \det \hat{H}_K(0) - \log \det \hat{H}_K(t). \quad (\text{CA35})$$

Subtract CA10 at the two ends of the same path. The result is exactly

$$\boxed{\overline{F}_K(t) = F_K(t) + \delta(t) - \delta(0)}. \quad (\text{CA36})$$

This proves IQR19 with its stated sign. With the opposite cost convention both cost signs would reverse; CA35 makes the convention explicit.

For four cutoff-dependent paths, each quantity $\delta_{r,N}(t_N) - \delta_{r,N}(0)$ can have either sign. Thus the max-of-two-sums argument in CA29 is applied to $\delta(t_N)$ itself, not directly to these differences. Applying CA26 instead gives the exact finite bound

$$|\mathcal{R}(\overline{F}_{K,N}(t_N) - F_{K,N}(t_N))| \leq \sum_{N=q-1, q, 2q-1, 2q} \ell_N \log(1 + t_N \|C_N\|). \quad (\text{CA37})$$

Under CA32 this is $o(kq)$, for any positive starting metric, without using CA27. Alternatively, when the actual starting metric also satisfies CA27, CA29 at time t_N and at zero, together with CA31–33, gives the same asymptotic conclusion. In particular the actual activated kernel cost and its quotient have the same kq coefficient when both costs refer to the same starting metric and the actual-baseline covariance obeys CA32.

The path holds K, V, π and their coefficient frames fixed. For a different variation in which π changes, direct differentiation gives

$$\partial(\pi G^{-1} \pi^*) = (\partial \pi) G^{-1} \pi^* + \pi G^{-1} (\partial \pi)^* - \pi G^{-1} (\partial G) G^{-1} \pi^*. \quad (\text{CA38})$$

Nothing in CA16 or CA36 removes the first two terms of CA38.

7.9 8. Exact examples testing sharpness, sign, and source cross terms

7.9.1 8.1 Rank one: sharp coefficient and changing cross-Gram rank

Take $E = \mathbb{C}^2$, $G_0 = I_2$, $K = \text{span}(e_1)$, $V = \text{span}(e_2)$, and

$$\Omega = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

This is a positive rank-one projection, so $C = \Omega$, $\|C\| = 1$. The identity $\Omega^2 = \Omega$ gives $G(t) = I_2 - \frac{t}{1+t} \Omega$. Its two diagonal entries are $(1 + t/2)/(1 + t)$, and its determinant is $(1 + t)^{-1}$. Hence

$$\delta(0) = 0, \quad \delta(t) = \log \frac{(1+t/2)^2}{1+t}. \quad (\text{CA39})$$

The quantity in the logarithm is at least 1 because $(1+t/2)^2 - (1+t) = t^2/4$. Its derivative is $\delta'(t) = t/((t+2)(t+1))$, which obeys CA25. Also

$$\delta(t) = \log t - \log 4 + o(1), \quad \frac{\delta(t)}{\log(1+t)} \rightarrow 1.$$

Thus the coefficient 1 in the rank-one form of CA26 cannot be replaced by a smaller universal coefficient. The cross Gram at zero is zero, but at $t > 0$ it is $-t/(2(1+t)) \neq 0$. Therefore $\nu(0) = 0$ cannot replace the uniform rank bound in CA26.

7.9.2 8.2 The angle loss need not increase

Let $0 < \rho < 1$, keep the same coordinate subspaces, and set

$$G_0 = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \Omega = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

In the orthonormal eigenvectors $(e_1 + e_2)/\sqrt{2}$, $(e_1 - e_2)/\sqrt{2}$, the metric eigenvalues are $a(t) = (1 + \rho)/(1 + t(1 + \rho))$ and $b = 1 - \rho$. Therefore

$$\delta(t) = \log \frac{(a(t) + b)^2}{4a(t)b}, \quad \delta(0) = -\log(1 - \rho^2) > 0. \quad (\text{CA40})$$

At $t_* = 2\rho/(1 - \rho^2)$, direct substitution gives $a(t_*) = b$, so $G(t_*) = bI_2$ and $\delta(t_*) = 0$. This exact decrease proves why the absolute-value estimate is needed and why a signed difference cannot automatically be treated as nonnegative.

7.9.3 8.3 Rank two with a full non-diagonal source Gram

Take $E = \mathbb{C}^4$, $G_0 = I_4$, $K = \text{span}(e_1, e_2)$, $V = \text{span}(e_3, e_4)$, and the complete two-column source

$$H_Z = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad F_Z = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here $H_Z^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and

$$\Omega = F_Z H_Z^{-1} F_Z^* = \frac{1}{2} \begin{pmatrix} H_Z^{-1} & H_Z^{-1} \\ H_Z^{-1} & H_Z^{-1} \end{pmatrix}. \quad (\text{CA41})$$

Every displayed off-diagonal entry is retained. On the subspace of vectors $(x, x)/\sqrt{2}$, the metric $G(t)$ acts as $A(t) = (I_2 + tH_Z^{-1})^{-1}$; on $(x, -x)/\sqrt{2}$, it is the identity. Therefore its diagonal coordinate blocks are $(I_2 + A(t))/2$, and $\det G(t) = \det A(t)$. The eigenvalues of H_Z^{-1} are 1, 1/3, so

$$\delta(t) = \log \frac{(1+t/2)^2}{1+t} + \log \frac{(1+t/6)^2}{1+t/3}. \quad (\text{CA42})$$

This verifies the complete-covariance interpretation with rank two. At $t = 1$, the answer is $\log(147/128)$. Deleting the cross terms in H_Z would instead replace it by $2I_2$ and give $\log(625/576)$, a different answer. Thus this finite example detects an unauthorized source-Gram truncation.

Finally, $\delta(t) = 2 \log t + O(1)$, so $\delta(t)/\log(1+t) \rightarrow 2$. The dimension/rank coefficient 2 in CA26 is also optimal in the class of rank-two examples. Direct sums of CA39 give the corresponding optimal coefficient for every positive integer rank allowed by the two subspace dimensions.

7.10 9. Complete coefficient-identity algebra

Retain the EIQ potential $\pi\sqrt{x} - 2\log x$, the positive square-root endpoints u, v , and the exact definitions from IQR2:

$$\begin{aligned} uK(\kappa) &= 2, & vE(\kappa) &= 4, \\ a &= u/v, & \kappa^2 &= 1 - a^2, & w &= (u+v)/2, & z &= uv, & c &= (v^2 - u^2)/4. \end{aligned} \quad (\text{CA43})$$

All logarithms below have positive arguments on the retained endpoint domain $0 < u < v$, $0 < \kappa < 1$. The letter c in this section is the original endpoint quantity $(v^2 - u^2)/4$, not the covariance norm in CA21.

The retained identities are

$$L = 6 \log w - 2 \log z - 2, \quad C_{\partial} = 4 \log(4/\pi) + 8 \log w - 4 \log z - 2 \log c, \quad (\text{CA44})$$

and

$$\psi(0) = \log(4/\pi), \quad \psi(1) = \log \frac{\kappa(1+a)}{E(\kappa)^2}, \quad J(1) = 2(\log 2 - 1) - L/2. \quad (\text{CA45})$$

The first value in CA45 is explicitly S3 in the cited retained proof. It is needed for the first displayed simplification in IQR2; the inspected receiver uses it without restating it.

From CA43, without rescaling either endpoint,

$$v = \frac{4}{E(\kappa)}, \quad w = \frac{v(1+a)}{2}, \quad \sqrt{c} = \frac{v\kappa}{2}.$$

Therefore

$$\frac{w\sqrt{c}}{4} = \frac{v^2\kappa(1+a)}{16} = \frac{\kappa(1+a)}{E(\kappa)^2}, \quad \psi(1) = \log w + \frac{1}{2} \log c - \log 4. \quad (\text{CA46})$$

Now expand every term:

$$\begin{aligned} 4[\psi(0) - \psi(1) - J(1) - 1] &= 4 \log(4/\pi) - 4 \log w - 2 \log c + 4 \log 4 \\ &\quad - 8 \log 2 + 8 + 2L - 4 \\ &= 4 \log(4/\pi) - 4 \log w - 2 \log c + 4 + 2L \\ &= 4 \log(4/\pi) - 4 \log w - 2 \log c + 4 \\ &\quad + 12 \log w - 4 \log z - 4 \\ &= 4 \log(4/\pi) + 8 \log w - 4 \log z - 2 \log c \\ &= C_{\partial}. \end{aligned} \quad (\text{CA47})$$

The cancellation uses $4 \log 4 = 8 \log 2$, and the additive constants $+4$ and -4 cancel only after the full L identity is inserted. Thus IQR2 has the correct constants, signs, and original endpoint scale. Its equality requires no numerical approximation. Any already proved interval for this same C_{∂} transfers through the equality, but the numerical interval itself has not been re-certified by this independent algebra check.

7.11 10. Precise conclusions for the receiver

- IQR15 is the exact fixed-period determinant differential in the original metric.
- IQR16 has the asserted paired spectrum; the vectors in its two-dimensional blocks belong to $G(t)^{1/2}K$ and $G(t)^{1/2}V$.
- IQR17 follows from CA26, which also gives the stronger sum over the top $\min(m, r, d)$ covariance eigenvalues.
- IQR18 is valid with the concatenated frame, the actual covariance ranks, the initial-angle estimate for the same starting metric, and the actual-baseline activated logarithmic-size bound CA32. It gives $o(kq)$ for ranks one and two under the stated $r \log(q+2) = o(q)$ regime when those hypotheses hold. An already activated baseline does not inherit the canonical IVO initial-angle estimate from rank alone.
- IQR19 is exact for the determinant-loss convention CA35. Its signed four-cutoff correction is controlled by CA37.
- IQR2 is algebraically correct, with the endpoint value $\psi(0) = \log(4/\pi)$ made explicit.

No counterexample occurs under these hypotheses. The exact examples identify three limits of stronger statements that were not asserted: the angle loss need not be monotone, the initial cross-Gram rank is not a uniform rank bound, and deleting source cross terms changes the answer.

Part IV

Complete supplied inverse-sector continuation

7.12 The inverse-power theorem gives a quantitative reduction of the original observation complex

The main propagated identity is

$$\mathcal{R} \log \det(I_K^* G_N I_K) = \mathcal{R} \log \det[(\pi_r I_K)^* (\pi_r G_N^{-1} \pi_r^*)^{-1} (\pi_r I_K)] + o_{h,A}(kq). \quad (1)$$

Here π_r is an explicit quotient by the first r inverse powers, defined below. The right side retains the original kernel, period, source measure, and full polynomial relation. It removes the large inverse-power directions without changing the kernel's kq coefficient.

At the same time, the removed observed sector carries the evaluated contribution

$$\mathcal{R} \log \det H_{r,N}^B = \mathfrak{C} r q + o_{h,A}(r q), \quad 1.353893 < \mathfrak{C} < 1.354751. \quad (2)$$

That contribution is transferred between specified determinant factors; it is not added to the original total while its compensating factor is held fixed.

The reduction also turns inverse multiplication into **ordinary Taylor division between successive quotient spaces**, with explicit subexponential norm bounds. This supplies a controlled arithmetic action after the reduction, rather than an unsupported assertion that multiplication descends to a fixed quotient.

7.12.1 Retained hypotheses and notation

Keep

$$q = (k + 1)^2, \quad m = \dim K = 8k - 16,$$

on the original simple-quartet and proved observation domain. Write

$$E = \mathbb{C}[y]/(Q_k), \quad M[p] = [yp],$$

$$\Lambda : E \longrightarrow B, \quad K = \ker \Lambda,$$

and retain the fixed original kernel frame I_K . At cutoff N ,

$$Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_{B,N} = G_N^{-1} \Lambda^* Q_{B,N}, \quad P_{B,N} = L_{B,N} \Lambda.$$

All coordinates use the original substitution $S = k/2 + iy$. A stored S -coefficient frame is carried to these coordinates by

$$(T_c)_{ab} = \binom{b}{a} c^{b-a} i^a, \quad c = k/2,$$

with

$$I_K^y = T_c I_K^S, \quad G_N^S = T_c^* G_N^y T_c.$$

Thus the phases are part of the coordinate map, not omitted from it.

Set

$$U_r d = \sum_{j=1}^r d_j [y^{-j}], \quad V_r = \text{im } U_r, \quad \Phi_r = \Lambda U_r,$$

$$H_{r,N}^E = U_r^* G_N U_r, \quad H_{r,N}^B = \Phi_r^* Q_{B,N} \Phi_r.$$

I use the preceding theorem in the following form:

$$\boxed{a_{r,N} H_{r,N}^E \preceq H_{r,N}^B \preceq H_{r,N}^E, \quad a_{r,N} \geq e^{-\varepsilon_{k,r}},} \quad (\text{IP})$$

where

$$\varepsilon_{k,r} = C_{h,A}(k + Jr) \log(q + Jr + 2) = o(q)$$

when

$$r \log(q + 2) = o(q).$$

Its explicit finite lower factor can be used instead of the asymptotic expression. J remains the number of nonzero original conductor coefficients.

The algebraic conductor and its full-kernel inclusion are the original IK1-IK8 identities; the source was read at the pinned repository version `fe22dc1dd51c633e82f93b731cbb6ec96b11ed64`. The coefficient limits continue to consume the stated monic-profile input, rather than an independently re-proved EIQ asymptotic.

The finite arguments below apply more generally to any positive finite source and onto observation satisfying (IP).

7.13 1. The original kernel loses only a controlled angle determinant

Fix one cutoff temporarily and suppress N . Put

$$H_K = I_K^* G I_K, \quad \Gamma = I_K^* G U_r.$$

Since $I - P_B$ is the original orthogonal kernel projection,

$$\boxed{H_r^B = H_r^E - \Gamma^* H_K^{-1} \Gamma.} \quad (3)$$

Now quotient the source by V_r . The image of K remains injective because $K \cap V_r = 0$, as proved by the conductor theorem. Its attained metric is

$$\boxed{\widehat{H}_{K,r} = H_K - \Gamma (H_r^E)^{-1} \Gamma^*.} \quad (4)$$

These two Schur complements have exactly the same nontrivial relative eigenvalues. Indeed, define

$$T = H_K^{-1/2} \Gamma (H_r^E)^{-1/2}.$$

Then

$$H_r^B = (H_r^E)^{1/2} (I - T^* T) (H_r^E)^{1/2},$$

$$\widehat{H}_{K,r} = H_K^{1/2} (I - T T^*) H_K^{1/2}.$$

Equation (IP) implies

$$T^*T \preceq (1 - a_{r,N})I.$$

Therefore

$$\boxed{a_{r,N}H_K \preceq \widehat{H}_{K,r} \preceq H_K.} \quad (5)$$

More importantly, the complete logarithmic loss is

$$\boxed{\begin{aligned} \delta_{r,N} &:= \log \det H_{r,N}^E - \log \det H_{r,N}^B \\ &= \log \det H_{K,N} - \log \det \widehat{H}_{K,r,N} \\ &= -\log \det(I - T^*T). \end{aligned}} \quad (6)$$

If $\nu_{r,N} = \text{rank } \Gamma$, only $\nu_{r,N}$ eigenvalues contribute:

$$\boxed{0 \leq \delta_{r,N} \leq \nu_{r,N} \varepsilon_{k,r} \leq \min(m, r) \varepsilon_{k,r}.} \quad (7)$$

The factor $\min(m, r)$, rather than r , matters when many inverse powers are retained. Because $m = O(k)$,

$$\boxed{\delta_{r,N} = o_{h,A}(kq)} \quad (8)$$

throughout the entire allowed regime $r \log q = o(q)$, uniformly over the original cutoffs.

For the four signs,

$$\boxed{\left| \mathcal{R} \log \det \widehat{H}_{K,r,N} - \mathcal{R} \log \det H_{K,N} \right| \leq 2 \min(m, r) \varepsilon_{k,r}.} \quad (9)$$

Each endpoint loss belongs to the same interval $[0, \min(m, r) \varepsilon_{k,r}]$; two occur with each sign. The factor two therefore retains the correlation of the four errors.

The estimate is sharp given only (IP). In an orthonormal space, take ν unit vectors of V_r of the form

$$\sqrt{1-a} e_j + \sqrt{a} f_j,$$

where $e_j \in K$ and $f_j \perp K$, and put the remaining vectors perpendicular to K . Then exactly ν relative eigenvalues equal a , and

$$\delta = \nu \log(1/a).$$

7.13.1 Explicit quotient coordinates

No numerical nullspace choice is needed for E/V_r . Define

$$\boxed{\pi_r[p] = \mathcal{D}^r(\text{rem}_{Q_k}(y^r p)), \quad \mathcal{D}^r f = \frac{f - \text{Tay}_{<r} f}{y^r}.} \quad (10)$$

Its target is the coefficient space $\mathcal{P}_{<q-r}$.

Multiplication by y is invertible modulo Q_k , and

$$M^r V_r = \mathcal{P}_{<r}.$$

Consequently

$$\ker \pi_r = V_r.$$

The ordinary low-degree inclusion

$$\iota_r : \mathcal{P}_{< q-r} \hookrightarrow E$$

satisfies

$$\pi_r \iota_r = I.$$

The actual quotient metric is

$$\boxed{\overline{G}_{r,N} = (\pi_r G_N^{-1} \pi_r^*)^{-1}}. \quad (11)$$

This is not the raw polynomial Gram on $\mathcal{P}_{< q-r}$. It is the minimum over all original representatives after quotienting by the specified inverse-power space.

The image of the original kernel has frame

$$\overline{I}_{K,r} = \pi_r I_K,$$

and

$$\overline{I}_{K,r}^* \overline{G}_{r,N} \overline{I}_{K,r} = \widehat{H}_{K,r,N}.$$

Equations (8)–(11) prove the opening identity (1).

For example, one may take

$$r_k = \left\lfloor \frac{q}{\log^2(q+2)} \right\rfloor.$$

Then r_k is much larger than m , but

$$\delta_{r_k,N} = O_{h,A} \left(\frac{kq}{\log q} \right) = o(kq).$$

Thus considerably more than $O(k)$ inverse directions can be removed while preserving the original kernel coefficient.

7.14 2. This is a chain-homotopy reduction, not merely a determinant comparison

Let

$$W_r = \Lambda V_r \subset B.$$

Choose a fixed algebraic quotient map

$$\rho_r : B \longrightarrow B/W_r.$$

It can be constructed from an actual nonzero $r \times r$ minor of Φ_r : retain those pivot rows and subtract their exact interpolation from the remaining rows. The matrix is fixed as N varies.

There is an induced onto map

$$\overline{\Lambda}_r : E/V_r \longrightarrow B/W_r, \quad \overline{\Lambda}_r \pi_r = \rho_r \Lambda. \quad (12)$$

Its kernel is exactly $\pi_r K$. The diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & V_r & \rightarrow & E & \xrightarrow{\pi_r} & E/V_r \rightarrow 0 \\ & & \downarrow \Lambda|_{V_r} & & \downarrow \Lambda & & \downarrow \bar{\Lambda}_r \\ 0 & \rightarrow & W_r & \rightarrow & B & \xrightarrow{\rho_r} & B/W_r \rightarrow 0 \end{array} \quad (13)$$

has exact rows and an isomorphism on the left.

The attained target metric is

$$\bar{Q}_{r,N} = (\rho_r Q_{B,N}^{-1} \rho_r^*)^{-1}.$$

Direct substitution proves

$$\boxed{\bar{Q}_{r,N} = (\bar{\Lambda}_r \bar{G}_{r,N}^{-1} \bar{\Lambda}_r^*)^{-1}}. \quad (14)$$

Thus either order of the two complete quotient minima gives the same metric.

7.14.1 An explicit contracting homotopy

Define

$$\boxed{h_{r,N} = U_r(H_{r,N}^B)^{-1} \Phi_r^* Q_{B,N} : B \rightarrow E}. \quad (15)$$

Then

$$\Lambda h_{r,N} = \Phi_r(H_{r,N}^B)^{-1} \Phi_r^* Q_{B,N}$$

is the $Q_{B,N}$ -orthogonal projection onto W_r , whereas $h_{r,N} \Lambda$ is the projection onto V_r along

$$\Lambda^{-1}(W_r^{\perp Q_{B,N}}).$$

The latter is generally oblique, not orthogonal.

Equation (IP) gives

$$\boxed{\|h_{r,N}\|_{Q_{B,N} \rightarrow G_N} \leq a_{r,N}^{-1/2} \leq e^{\varepsilon_{k,r}/2}}. \quad (16)$$

For the finite cochain complex

$$\mathcal{C}_N^\bullet = [E \xrightarrow{\Lambda} B],$$

the two projections

$$I - h_{r,N} \Lambda, \quad I - \Lambda h_{r,N}$$

give its complementary subcomplex, and

$$I - jp = dh + hd. \quad (17)$$

Here $p = (\pi_r, \rho_r)$; an explicit chain section j is obtained by composing the minimum section of π_r with $I - h\Lambda$, and the minimum section of ρ_r in degree one.

This proves a chain-homotopy equivalence

$$\boxed{[E \rightarrow B] \simeq [E/V_r \rightarrow B/W_r]}. \quad (18)$$

Its degree-zero cohomology remains the original observation kernel K . This is a statement about this finite observation complex, not an unconstructed identification with the entire infinite theta complex. The standard meanings of “acyclic” and “quasi-isomorphism” are as in the Stacks Project; the homotopy itself is explicitly given in (15)–(17). (The Stacks Project)

7.14.2 The full polynomial-source relations survive

Let

$$J_N : \mathcal{P}_N \rightarrow E, \quad L_N : E \rightarrow \mathcal{P}_N$$

be the actual remainder map and its original minimum section. The physical polynomial lift of the contracting homotopy is

$$\tilde{h}_{r,N} = L_N h_{r,N}.$$

Its range is the actual source subspace $L_N V_r$, and

$$\|\tilde{h}_{r,N}\| = \|h_{r,N}\|.$$

The complete relation space

$$Q_k \mathcal{P}_{N-q}$$

is untouched: J_N kills it, so the contraction vanishes on it. It is also orthogonal to $L_N V_r$. Hence the same reduction lifts to

$$[\mathcal{P}_N \xrightarrow{\Lambda J_N} B]$$

without deleting any original polynomial relation.

7.15 3. The determinant contribution moves with a known coefficient

There are two different operations.

If only the target is quotiented by W_r , the new kernel is

$$K_r = K + V_r.$$

In the fixed frame $[I_K, U_r]$,

$$\boxed{\det([I_K, U_r]^* G_N [I_K, U_r]) = \det H_{K,N} \det H_{r,N}^B.} \quad (19)$$

This is the Schur complement (3).

If both source and target are quotiented, as in (13), the kernel remains isomorphic to K , with the controlled metric change (6).

For the target-only quotient, choose one fixed complement to W_r in B . Its coefficient-frame determinant is independent of N , so

$$\boxed{\mathcal{R} \log \det \bar{Q}_{r,N} = \mathcal{R} \log \det Q_{B,N} - \mathcal{R} \log \det H_{r,N}^B.} \quad (20)$$

Suppose $r/k \rightarrow \lambda$. The preceding theorem therefore gives the exact leading redistribution

$$\boxed{\begin{aligned} \frac{\mathcal{R} \log \det G_{K_r,N} - \mathcal{R} \log \det G_{K,N}}{kq} &\longrightarrow \lambda \mathfrak{C}, \\ \frac{\mathcal{R} \log \det \overline{Q}_{r,N} - \mathcal{R} \log \det Q_{B,N}}{kq} &\longrightarrow -\lambda \mathfrak{C}. \end{aligned}} \quad (21)$$

The two contributions cancel in the total determinant identity. For $r = k$, the transferred coefficient is precisely $\mathfrak{C}$.

Meanwhile the relative squared-volume cost of the contractible block

$$V_r \xrightarrow{\Lambda} W_r$$

is

$$\log \det H_{r,N}^E - \log \det H_{r,N}^B = \delta_{r,N} = o(kq).$$

Thus the large, evaluated $+\mathfrak{C}r q$ term occurs in both its source and observed volumes; their relative distortion is subleading.

7.16 4. The arithmetic inverse becomes bounded Taylor division on the reduced spaces

The quotient in (18) is not automatically stable under multiplication by M . The following gives the exact replacement.

Let

$$\mathscr{D}^s p = \frac{p - \text{Taylor}_{<s} p}{y^s}.$$

For $x \in E$, write the actual minimum representative as

$$L_N x = \sum_{j=0}^{s-1} c_j(x) y^j + y^s \mathscr{D}^s L_N x.$$

Then

$$\boxed{M^{-s} = J_N \mathscr{D}^s L_N + U_s F_{s,N}, \quad (F_{s,N} x)_n = c_{s-n}(x).} \quad (22)$$

These c_j are divided Taylor coefficients of $L_N x$, not evaluation functionals asserted to descend directly to E .

Retain

$$a_N^\Gamma = \sqrt{n_N(n_N + 1)}(1 + \sqrt{N + 1}), \quad n_N = \left\lfloor \frac{N + 2}{2} \right\rfloor.$$

The supplied Gamma division estimate and native comparison give

$$\boxed{\|J_N \mathscr{D}^s L_N\| \leq \mathfrak{B}_{s,N} := \sqrt{\frac{u_k}{\ell_{k,N}}} (a_N^\Gamma)^s.} \quad (23)$$

Only one native/Gamma comparison is used; it is not multiplied s times. In particular,

$$\log^+ \mathfrak{B}_{s,N} = O_h(k + s \log(q + 2))$$

on the relevant degree range. The retained division estimate and its mass convention are S11–S16 of the supplied proof.

7.16.1 Exact degree-changing inverse maps

For $r + s \leq q$, multiplication by M^{-s} induces

$$\mathcal{I}_{r,s} : E/V_r \longrightarrow E/V_{r+s}, \quad [x] \longmapsto [M^{-s}x].$$

It is well-defined because $M^{-s}V_r \subset V_{r+s}$.

In the canonical polynomial coordinates (10),

$$\boxed{\mathcal{I}_{r,s}p = \mathcal{D}^s p.} \quad (24)$$

Indeed, for $\deg p < q - r$, the polynomial $y^r p$ already has degree below q . Applying (10) to $M^{-s}p$ therefore drops exactly the first s coefficients of p .

The whole sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_{<s} \longrightarrow E/V_r \xrightarrow{\mathcal{I}_{r,s}} E/V_{r+s} \longrightarrow 0.} \quad (25)$$

To check the kernel, multiply by M^s :

$$M^s V_{r+s} = V_r + \mathcal{P}_{<s}.$$

The sum is direct because multiplication by M^r sends its two parts to disjoint monomial ranges of degree below q .

Equation (22) gives the original-metric bound

$$\boxed{\|\mathcal{I}_{r,s}\|_{\bar{G}_{r,N} \rightarrow \bar{G}_{r+s,N}} \leq \mathfrak{B}_{s,N}.} \quad (26)$$

The large rank- s term in (22) lands in $V_s \subset V_{r+s}$ and vanishes in this specified quotient.

These maps compose exactly:

$$\mathcal{I}_{r+s,t} \mathcal{I}_{r,s} = \mathcal{I}_{r,s+t}. \quad (27)$$

7.16.2 The observation retains a complete memory term

Let $\bar{L}_{r,N}$ be the minimum section of $\bar{\Lambda}_r$. Define the observed inverse-step map

$$\mathcal{B}_{r,s,N} = \bar{\Lambda}_{r+s} \mathcal{I}_{r,s} \bar{L}_{r,N}.$$

Then

$$\boxed{\|\mathcal{B}_{r,s,N}\| \leq \mathfrak{B}_{s,N}.} \quad (28)$$

It is not asserted to be the inverse of an independently compressed observed multiplication operator.

Let $\bar{P}_{r,N}^K$ be the projection onto $\ker \bar{\Lambda}_r$. The full action identity is

$$\boxed{\bar{\Lambda}_{r+s} \mathcal{I}_{r,s} = \mathcal{B}_{r,s,N} \bar{\Lambda}_r + \bar{\Lambda}_{r+s} \mathcal{I}_{r,s} \bar{P}_{r,N}^K.} \quad (29)$$

The second term is the retained kernel leakage, with norm at most $\mathfrak{B}_{s,N}$. Likewise,

$$\boxed{\mathcal{B}_{r,s+t,N} - \mathcal{B}_{r+s,t,N} \mathcal{B}_{r,s,N} = \bar{\Lambda}_{r+s+t} \mathcal{I}_{r+s,t} \bar{P}_{r+s,N}^K \mathcal{I}_{r,s} \bar{L}_{r,N}.} \quad (30)$$

Its rank is at most m , and its norm is at most

$$\mathfrak{B}_{t,N} \mathfrak{B}_{s,N}.$$

Thus the observation's failure to compose is not discarded; it is a specified, bounded kernel-feedback map.

For the original kernel itself, (22) gives the particularly direct identity

$$\boxed{\rho_{r+s} \Lambda M^{-s} I_K = \rho_{r+s} \Lambda J_N \mathcal{D}^s L_N I_K,} \quad (31)$$

whose norm, from $H_{K,N}$ to the attained target metric, is at most $\mathfrak{B}_{s,N}$. The full inverse has norm of order

$$\exp[q \log(q/k) + O_h(q)].$$

After the specified quotient, the actual kernel-to-observation inverse map has norm $\exp(o(q))$ whenever $s \log q = o(q)$.

7.16.3 A full shift-and-minimize realization of the original kernel

There is a useful version involving only original polynomial source Grams.

Let $r \log q = o(q)$. In $\mathcal{P}_{N+r}$, form the complete relation matrix whose columns are

$$1, y, \dots, y^{r-1}, \quad Q_k, yQ_k, \dots, y^{N+r-q}Q_k.$$

Call it $R_{r,N}$. Let $E_{r,N}$ contain the coefficient columns of $y^r I_K$, using the original degree-below- q representatives.

With the full source Gram H_{N+r}^{src} , define

$$\boxed{\begin{aligned} \mathcal{H}_{r,N}^{\text{shift}} &= E_{r,N}^* H_{N+r}^{\text{src}} E_{r,N} \\ &\quad - E_{r,N}^* H_{N+r}^{\text{src}} R_{r,N} (R_{r,N}^* H_{N+r}^{\text{src}} R_{r,N})^{-1} R_{r,N}^* H_{N+r}^{\text{src}} E_{r,N}. \end{aligned}} \quad (32)$$

Every inverse is of a complete positive Gram. Both the low-polynomial directions and all shifted Q_k -relations remain.

Multiplication by y^r gives the exact isomorphism

$$E/V_r \longrightarrow E/\mathcal{P}_{<r}.$$

The source cutoff is N ; the target cutoff is $N+r$.

A forward norm bound is

$$\mathfrak{A}_{r,N} = \prod_{j=0}^{r-1} C_{\text{mult}}(h) [2(N+j) + \lfloor k/3 \rfloor].$$

The inverse bound is $\mathfrak{B}_{r,N+r}$. Indeed, multiply a minimum source representative by y^r for the forward estimate; for the inverse, apply $\mathcal{D}^r$ to a complete target representative. Its discarded Taylor polynomial is precisely an allowed target relation.

Consequently,

$$\boxed{\mathfrak{B}_{r,N+r}^{-2} \widehat{H}_{K,r,N} \preceq \mathcal{H}_{r,N}^{\text{shift}} \preceq \mathfrak{A}_{r,N}^2 \widehat{H}_{K,r,N}.} \quad (33)$$

Since

$$\log^+ \mathfrak{A}_{r,N} + \log^+ \mathfrak{B}_{r,N+r} = O_h(k + r \log q),$$

equations (7) and (33) prove

$$\boxed{\mathcal{R} \log \det(I_K^* G_N I_K) = \mathcal{R} \log \det \mathcal{H}_{r,N}^{\text{shift}} + o_{h,A}(kq).} \quad (34)$$

Equation (34) is a new explicit polynomial-source target for the remaining kernel coefficient. Its four target cutoffs are $N+r$; they have not been relabelled as the old cutoffs.

7.17 5. The late heat is localized in the removed observed sector

The source identifies

$$u = -Q_k(0)[y^{-1}]$$

as the surviving late-heat line and proves, on its finite guard,

$$\|P_{\min,N} - P_{u,N}\| \leq \eta_N, \quad \eta_N = \frac{D_N}{E_N - D_N}.$$

The remaining singular values are at least D_N^{-1} . These are the original LS1-LS5 statements, with the original metric.

Let

$$W_{r,N}^E = P_{B,N} V_r,$$

and let $P_{W,r,N}$ be its original-metric orthogonal projection:

$$P_{W,r,N} = P_{B,N} U_r (H_{r,N}^B)^{-1} U_r^* G_N P_{B,N}.$$

Then

$$P_{r,N}^{\text{rem}} = P_{B,N} - P_{W,r,N}$$

is the projection onto $(K + V_r)^{\perp_{G_N}}$.

Since $u \in V_1 \subset V_r$,

$$P_{r,N}^{\text{rem}} u = 0.$$

If a_N is the unit smallest right singular vector,

$$\|P_{r,N}^{\text{rem}} a_N\|^2 \leq \eta_N^2.$$

Therefore

$$\boxed{0 \leq \text{Tr}(P_{r,N}^{\text{rem}} e^{-\tau M^\dagger M}) \leq \eta_N^2 e^{-\tau s_{\min,N}^2} + (q-1)e^{-\tau/D_N^2}.} \quad (35)$$

The error is quadratic in the line-angle bound here because the residual projection kills the reference line exactly.

The preceding inverse-power theorem gives

$$\theta_N = \frac{\|P_{B,N} u\|^2}{\|u\|^2} \geq e^{-O_{h,A}(k \log q)}.$$

Meanwhile

$$\eta_N = \exp[-q \log(q/k) + O_h(q)].$$

It follows that, at the same common late time

$$\tau_k^* = \exp \left\{ 2q \left[\log(q/k) - 1 - \frac{I(\delta, \gamma)}{2} \right] \right\},$$

the original measured heat is asymptotically entirely carried by $W_{r,N}^E$:

$$\boxed{\frac{\text{Tr}(P_{W,r,N} e^{-\tau_k^* M^\dagger M})}{\text{Tr}(P_{B,N} e^{-\tau_k^* M^\dagger M})} \rightarrow 1.} \quad (36)$$

For the four-cutoff return, the residual contribution obeys

$$\boxed{\limsup_{k \rightarrow \infty} \frac{\log \left| \mathcal{R} \text{Tr}(P_{r,N}^{\text{rem}} e^{-\tau_k^* M^\dagger M}) \right|}{q \log(q/k)} \leq -2,} \quad (37)$$

with $\log 0 = -\infty$.

Thus the positive late-heat event from the preceding theorem is located in a specific contractible observed sector. Its determinant contribution is already accounted for by (19)–(21).

7.17.1 The remaining positive square has a finite gap

Let $S_{r,N} : E/V_r \rightarrow E$ be the original minimum section of π_r , with image $V_r^\perp$. For $x \perp V_r$,

$$x = (I - P_{V_r})M^{-1}Mx = (I - P_{V_r})J_N \mathcal{D} L_N Mx.$$

Hence

$$\|Mx\| \geq D_N^{-1} \|x\|.$$

The full-output positive operator therefore satisfies

$$\boxed{S_{r,N}^\dagger M^\dagger M S_{r,N} \succeq D_N^{-2} I.} \quad (38)$$

This is the positive square of the rectangular map $MS_{r,N}$. It is not identified with either the square of a compressed endomorphism or the compression of its exponential.

7.17.2 Applying the original arithmetic action preserves the kernel coefficient

Put

$$Z_{r,N} = (I - P_{V_{r,N}})I_K, \quad \mathcal{E}_{K,r,N}^{(s)} = (M^s Z_{r,N})^* G_N (M^s Z_{r,N}),$$

where $1 \leq s \leq r$.

Equation (22) gives

$$\|M^s x\| \geq \mathfrak{B}_{s,N}^{-1} \|x\|, \quad x \perp V_r.$$

For the upper determinant bound, the rank-one relation $M = C_N + R_N$ implies

$$\sigma_j(M) \leq B_h q \quad (j \geq 2), \quad \sigma_1(M) \leq \epsilon_N + B_h q.$$

Exterior submultiplicativity therefore gives

$$\begin{aligned}
-2m \log \mathfrak{B}_{s,N} &\leq \log \frac{\det \mathcal{E}_{K,r,N}^{(s)}}{\det \widehat{H}_{K,r,N}} \\
&\leq 2s \log(\epsilon_N + B_h q) + 2s(m-1) \log(B_h q).
\end{aligned} \tag{39}$$

Only one large singular value of M enters the exterior bound; it is not charged independently in all m directions.

Using the retained $\log \epsilon_N = O_h(q)$, for $s = o(k)$,

$$\mathcal{R} \log \det \mathcal{E}_{K,r,N}^{(s)} = \mathcal{R} \log \det(I_K^* G_N I_K) + o_{h,A}(kq). \tag{40}$$

The kernel coefficient is therefore preserved both by the quantified inverse-sector quotient and by applying these powers of the original arithmetic action to its minimum representatives.

7.18 6. Two existing programme terms can now be sharpened

7.18.1 The eigenclass angle term in NP19 is $o(q)$

For a conductor-detected original eigenclass v_λ , the preceding theorem gives

$$\beta_N(\lambda) = \frac{\|\Lambda v_\lambda\|_{Q_{B,N}}^2}{\|v_\lambda\|_{G_N}^2} \geq e^{-C_{h,A} k \log(q+2)}.$$

Therefore

$$|\mathcal{R} \log \beta_N(\lambda)| \leq 2C_{h,A} k \log(q+2) = o(q). \tag{41}$$

This closes the leading-order uncertainty in NP19's displayed canonical observation-angle correction on the conductor-detected eigenline domain. The source's complete output-stack identity is NP18; it retains that angle rather than removing it by observability alone.

In fact, for any depth $d = d(k)$ used identically at all four cutoffs,

$$\|\mathcal{O}_d v_\lambda\|^2 = \left(\sum_{j=0}^d |\lambda|^{2j} \right) \|\Lambda v_\lambda\|^2.$$

The entire geometric factor cancels under $\mathcal{R}$, even when d grows. Thus

$$\mathcal{R} \log \|\mathcal{O}_d v_\lambda\|^2 = \mathcal{R} \log \|v_\lambda\|_{G_N}^2 + O_{h,A}(k \log q). \tag{42}$$

For the unchanged cardinal eigenclass of the preceding calculation, the q -coefficient is therefore $\mathfrak{C}$.

The physical class must remain fixed as N varies; separate unit normalization at each cutoff would change this statement. Equations (41)–(42) concern squared norms, not the sign of the different complex product $\Re(\bar{z}w)$.

7.18.2 The full inverse-power current has an $O(k)$ -dimensional receiver

The orthogonal decomposition

$$E = K \oplus W_{r,N}^E \oplus (K + V_r)^\perp$$

has the exact block-diagonal metric

$$H_{K,N} \oplus H_{r,N}^B \oplus \overline{Q}_{r,N}$$

in the corresponding minimum-section frame.

For $x = U_r d$, its last component is zero, and its first two coordinates are

$$\kappa = H_K^{-1} \Gamma d, \quad d.$$

Let A be the original arithmetic action, and retain its actual blocks in this frame. Then the original pair is exactly

$$\boxed{z = \kappa^* H_K A_{K,W} d, \quad w = d^* H_r^B A_{W,K} \kappa.} \quad (43)$$

Thus the current on the entire inverse-power sector is computed in dimension $m + r$, with both cross blocks and their phases retained. For $r = k$, this is $9k - 16$, rather than $q = (k + 1)^2$.

Higher powers require the full return through the third block. For example,

$$P_{K+V_r} A^2 P_{K+V_r} = (P_{K+V_r} A P_{K+V_r})^2 + P_{K+V_r} A P_{\text{rem}} A P_{K+V_r}. \quad (44)$$

The last term is not discarded. This is the same distinction enforced by the full moving-section action formula M16 in the supplied manuscript.

7.19 7. The reduction commutes with the complete mixed-source covariance

The general-family source supplies the actual normal Gram and value map

$$H_Z, \quad F_Z, \quad \ker H_Z \subseteq \ker F_Z,$$

and

$$\Omega_N = F_Z H_Z^+ F_Z^*.$$

Its source-activation metric is

$$G_N(t) = (G_N^{-1} + t\Omega_N)^{-1}.$$

These maps contain the full cross Gram between the scalar source and the added tensor section.

Quotienting by the fixed V_r gives, exactly,

$$\boxed{\overline{G}_{r,N}(t) = [\overline{G}_{r,N}(0)^{-1} + t\pi_r \Omega_N \pi_r^*]^{-1}.} \quad (45)$$

The new normal value map is simply $\pi_r F_Z$, with the same full H_Z^+ . The compatibility of its kernel follows from the original one.

Let

$$\mathcal{C}_N = G_N^{1/2} \Omega_N G_N^{1/2}.$$

The covariance on E/V_r , in its initial metric, is a compression

$$\bar{\mathcal{C}}_{r,N} = S_r \mathcal{C}_N S_r^*, \quad S_r S_r^* = I_{q-r}.$$

Interlacing gives

$$0 \leq \text{Tr} \frac{t \mathcal{C}_N}{I + t \mathcal{C}_N} - \text{Tr} \frac{t \bar{\mathcal{C}}_{r,N}}{I + t \bar{\mathcal{C}}_{r,N}} \leq r, \quad (46)$$

and

$$0 \leq \log \det(I + t \mathcal{C}_N) - \log \det(I + t \bar{\mathcal{C}}_{r,N}) \leq r \log(1 + t \|\mathcal{C}_N\|). \quad (47)$$

Consequently the supplied effective-dimension law

$$q^{-1} \text{Tr} \frac{t_k(b) \mathcal{C}_N}{I + t_k(b) \mathcal{C}_N} \rightarrow (1 - b)_+, \quad t_k(b) = e^{-2bq \log(q/k)},$$

survives this quotient, with an additional finite error at most r/q . The original law and its precise normalization are C6-C10.

Likewise, the source's bound

$$\log(1 + \|\mathcal{C}_N\|) = O_h(q \log(q/k))$$

and $r \log q = o(q)$ show that the quotient changes its full metric cost by $o(q^2)$. Thus the known weak-activation return remains

$$\mathcal{R} \log \det(I + e^{-2\theta q} \bar{\mathcal{C}}_{r,N}) = C_B q^2 + o_h(q^2). \quad (48)$$

The same conclusion holds after the additional original observation quotient, whose codimension contributes another $m = o(q/\log q)$. This uses the supplied H18-H23 domain, not a new evaluation of its original profile.

At the finer kernel scale, there remains an exact angle correction. Define $\delta_{r,N}(t)$ by (6), now using $G_N(t)$. If

$$F_K(t) = \log \det H_K(0) - \log \det H_K(t),$$

and $\bar{F}_K$ is the corresponding deflated quantity, then

$$\bar{F}_K(t) = F_K(t) + \delta_{r,N}(t) - \delta_{r,N}(0). \quad (49)$$

The canonical bound controls $\delta_{r,N}(0)$. It does not assign $\delta_{r,N}(t)$ after a large source change.

For a varying period, rather than a fixed-packet source activation, the quotient map itself moves. Its complete covariance derivative is

$$\begin{aligned} \partial(\pi_r G^{-1} \pi_r^*) &= (\partial \pi_r) G^{-1} \pi_r^* + \pi_r G^{-1} (\partial \pi_r)^* \\ &\quad - \pi_r G^{-1} (\partial G) G^{-1} \pi_r^*. \end{aligned} \quad (50)$$

Thus period derivatives do not lose the two quotient-frame terms.

7.20 8. The proper-source comparison retains its entire target minimum

The independent proper-source matrix remains

$$H_e^p = F_{R,e}^* \mathcal{D}_e^p F_{R,e} - F_{R,e}^* \mathcal{D}_e^p J_* (J_*^* \mathcal{D}_e^p J_*)^{-1} J_*^* \mathcal{D}_e^p F_{R,e}. \quad (51)$$

Here e indexes its four original endpoints. The full J_* , the independent source metric Q_e^p , and the original degree shifts remain. These are precisely the NP30–NP31 objects.

Let their common input dimension be $r_p = O(k)$. Use the ordered inverse-power frame of length r_p only as an explicit comparison space, at

$$\hat{N}_e = q - 1, \quad q, \quad 2q - 1, \quad 2q$$

respectively. This does not identify $\hat{N}_e$ with the proper-source degrees. There are exact maps

$$U_{r_p} d \mapsto d$$

into the original proper-source coordinate frame, and

$$\Phi_{r_p} d \mapsto [F_{R,e} d] \pmod{\text{im } J_*}$$

into the full target quotient. The second is well-defined because Φ_{r_p} is injective. It makes the corresponding source–observation square commute.

Taking determinants in that common coefficient frame gives

$$\mathcal{M}^p = \mathcal{R}_e \left[\log \frac{\det H_e^p}{\det H_{r_p, \hat{N}_e}^B} - \log \frac{\det Q_e^p}{\det H_{r_p, \hat{N}_e}^E} \right] - \mathcal{R} \delta_{r_p, N}. \quad (52)$$

Since $\mathcal{R} \delta_{r_p, N} = o(kq)$,

$$\mathcal{M}^p = \mathcal{R}_e \left[\log \frac{\det H_e^p}{\det H_{r_p, \hat{N}_e}^B} - \log \frac{\det Q_e^p}{\det H_{r_p, \hat{N}_e}^E} \right] + o(kq). \quad (53)$$

This is the exact propagation into the proper-source branch. The common evaluated inverse-power coefficient cancels between its source and target reference volumes. The remaining two comparison determinants contain the actual proper-source geometry and its full target Schur term; they have not been assigned values by the cancellation.

7.21 What has changed in the programme

The inverse-power result now supplies more than another nonvanishing theorem.

The original observation kernel has a **cohomology-preserving, quantitatively equivalent quotient realization** at the kq scale, equations (1), (18), and (34). The known inverse-sector coefficient is allocated exactly between kernel augmentation and residual observation, equation (21). The previously unresolved canonical eigen-class angle is now $o(q)$ on the conductor-detected domain, equation (41).

The arithmetic inverse on the reduced spaces is the explicit Taylor-division sequence (24)–(27), with the complete observed feedback term (30). The late-heat event is isolated in the removed observed sector with the stronger residual bound (35). The large mixed-source covariance law survives, while its finer kernel correction is identified explicitly by (49).

The remaining fixed-kernel coefficient may therefore be calculated from the full shifted-source matrix (32), with a proved $o(kq)$ error. The independent proper-source coefficient is the exact comparison in (53). Neither step requires replacing the original kernel by a generically oriented subspace or deleting a target-invariant direction.

No executable verification completed in this turn: the runtime calls timed out. The added statements above are written proofs from the preceding theorem and the cited source identities; no repository files were changed.

7.21.1 References

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